ME 1100 : Thermodynamics

Class notes

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Chapter 1 Fundamentals

1.1 Overview of Thermodynamics

Industrial Revolution brought in large scale automation of many tedious tasks which were earlier being performed through manual or animal labour. Inventors discovered many machines which could be operated with steam power or run with combustion gases generated from the burning of fossil fuels. The possibility of converting energy from one form to another- particularly from heat to useful work was understood; thus was born the science of Thermodynamics. While the overall balance of energy in any process gave rise to the I Law of Thermodynamics, limitations on the performance indices of machines dealing with heat & work gave rise to the II law of Thermodynamics. Although Thermodynamics developed as a by-product of industrial revolution, in course of time, its application extended to the study of many physical and chemical processes. In particular, the changes that occur in the properties of substances during energy exchanges were characterized in detail. Also, the feasibility of any process under given conditions could be systematically studied. An amazing aspect of this subject is that its concepts could be applied to reactions occurring between tiny molecules or to gross changes which occur over the whole universe. The three basic laws of Thermodynamics provide very powerful tools to analyze the operation of virtually every machine or process on which our modern civilization is based. The only notable exceptions for classical thermodynamic analyses, one may say, are the processes that involve mass-to-energy conversion as in the case of nuclear particle interactions.

1.2 Basic Definitions

In any thermodynamic analysis, the very first task taken up is that of identifying the object or the spatial domain which forms the focus of the study. Two types of analyses are commonly performed- those corresponding to that of a **'System'** and that of a **'Control Volume'**. Some text books refer to the system as a **'Closed System'** and the control volume as an **'Open System'**.

A **System** is defined as a fixed mass of matter on which attention is focussed. The choice of the system is indicated with the help of a dash line, drawn very close to the boundary of the system on the interior side. The rest of the universe which lies outside the system is called the **'Surroundings'**. Let us consider the situation corresponding to the conversion of heat to work by a system, as shown in Fig. 1.1. Here, **Q** is the heat transferred from the surroundings to the system. Also, **W** is the work done by the system on surroundings. The difference between Q and W (say, equal to Δ**E**) is the change in the energy stored within the system.

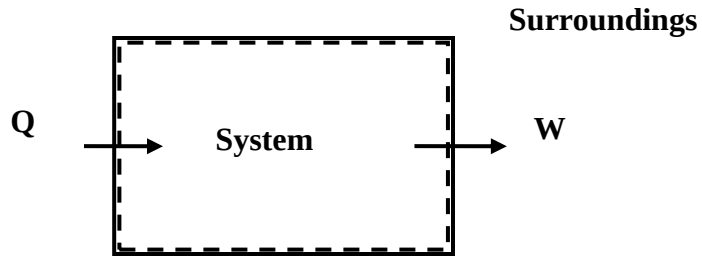


Fig. 1.1 Definition of System

Examples of system are: electric bulb, photovoltaic solar cell, nuclear power plant, domestic refrigerator, electrical heater, gas enclosed in a cylinder-piston device which expands on heating, an iron rod which is heated in a furnace. Note that in the electrical devices mentioned here, the electrical energy input or output is considered as work. More discussion about 'Work' is provided in the next chapter. Also, every process need not always have heat or work exchange- in some processes either Q or W (or both) could be zero.

A **'Control Volume'** (or CV for short) is a fixed volume on which attention is focussed. Apart from energy interactions in the form of heat or work, it can also admit in-flow or out-flow of mass through an inlet or an exit, respectively. Similar to the case of system, the rest of the universe outside the control volume is also called as **'Surroundings'**. The energy and mass interactions of a typical control volume are shown in Fig. 1.2. In the figure, Q and W represent the energy interactions and, m_i and m_e represent the in-flow and out-flow mass exchanges between the CV and the surroundings.

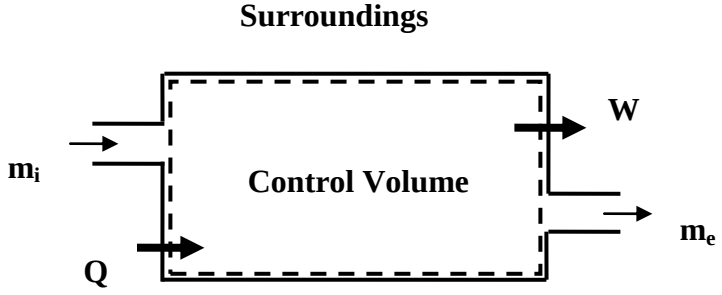


Fig. 1.2 Definition of CV

Examples of control volume are: water pump, air- conditioner, aircraft engine. It is evident from the above discussions that a system analysis permits only energy exchanges. It is possible that the volume of the system undergoes change during the process under consideration. For instance, the volume of gas enclosed in a cylinder- piston arrangement may change because of heat addition and this process can be modeled through the system analysis. In a control volume analysis, on the other hand, the volume of CV remains fixed although the mass enclosed inside CV could change. As in the case of the aircraft engine, the whole CV could be moving in some applications. A water tank which is being filled with water flowing from a tap could be analyzed as a CV- even when the drain of the tank is closed. Note that in this case Q and W are zero and $m_e = 0$. Thus, a control volume analysis can be applied when at least one of the two- namely the inlet or the outlet- is open and mass exchange occurs between the CV and the surroundings. There are some cases which can not be categorized either as a system or as a control volume. For instance, a balloon which is inflated with the air blown into it, is neither a system nor a CV. In such problems, it is better to fix a quantity of air (say, the air which finally occupies the balloon) and apply system analysis to this air.

Apart from the definitions of system, control volume and surroundings, there are a few other definitions which are fundamental to any thermodynamic analysis. These definitions are given below.

Property : Any measurable, macroscopic characteristic of a substance. Examples are : mass, volume, density, pressure, temperature, velocity, elevation, specific heat. Some quantities which may not be directly measurable but can be expressed in terms of measurable quantities are also treated as properties. For instance, kinetic energy is a property which can be expressed in terms of the mass and velocity. Potential energy in a gravitational field can be expressed in terms of the mass and elevation from a datum. Internal energy can be expressed in terms of the mass, specific heat and temperature. Properties can classified as intensive properties and extensive properties. An intensive property is independent of the size of the system (eg. density, pressure, temperature) while an extensive property depends on the size of the system (eg. mass, kinetic energy, volume).

State : 'State' represents the collection of all properties of a system. Consider for example, a system undergoing some process. Initially, the system properties may be: pressure p_1, volume V_1, temperature T_1, velocity V_1 and elevation z_1. All these properties could be taken together to define the **initial state '1'**. Similarly, if the final properties at the end of the process are given by: pressure p_2, volume V_2, temperature T_2, velocity V_2 and elevation z_2. These properties could

be taken together to define the **final state '2'**. In the case of some simple systems, two properties are often sufficient to define the state. For example, the state of a gas of given mass undergoing an expansion or compression process (with no motion or change in elevation), can be defined in terms of the pressure and volume. It is then possible to denote the initial and final states of the system as the points (p_1,V_1) and (p_2,V_2) in p-V diagram.

Process : A process is the sequence of states followed by a system. If a system changes from the initial state (p_1,V_1) to the final state (p_2,V_2), the curve connecting all the intermediate values of (p,V) for the system, represents the process undergone by the system in a p-V diagram.

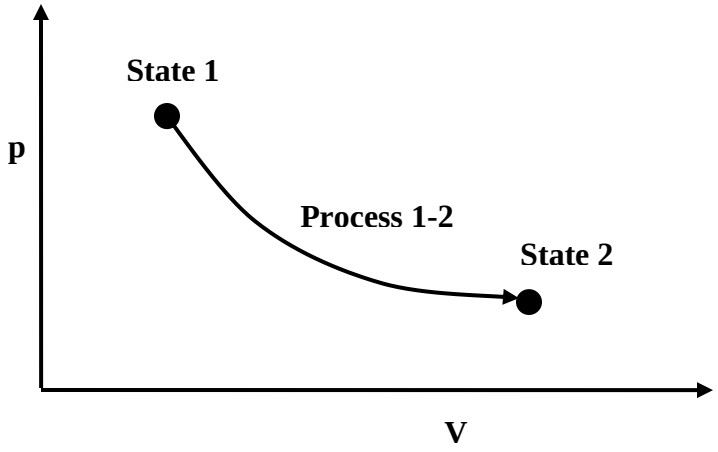


Fig. 1.3 State and Process

State dependent and path (process) dependent quantities :

When a system undergoes a process, there are many associated changes. For instance, the pressure, volume and temperature of the system may change. Some heat and work interactions may occur between the system and the surroundings. There is a need to quantify such changes- from the beginning to the end of the process, or some times even during the course of the process itself.

Let us consider a gas expansion process from state 1 (initial) to state 2 (final). Let the changes in pressure and volume during a small part of the process be: p-dp/2 to p+dp/2 and from V-dV/2 to V+dV/2 (Fig. 1.4 a). For the expansion process under consideration, the volume will expand (dV>0) while the pressure will decrease (dp<0). The overall change in pressure or volume can be obtained as:

$$\Delta p = \int_{p_1}^{p_2} dp = p_2 - p_1. \quad \Delta V = \int_{V_1}^{V_2} dV = V_2 - V_1.$$

It is evident from the above expressions that that the overall changes in pressure or volume depend only on the corresponding initial and final values of pressure or volume. They do not depend on the particular process- for example, both process x and process y shown in Fig. 1.4 b will have same Δp and ΔV. On the other hand, a quantity such as the work done will depend on the process. We will show in the next chapter that the work done during a small change in volume and pressure is given by p.dV where p is the average pressure and dV is the change in volume, as shown in Fig. 1.4 a. Work done during gas expansion is the shaded area shown in the figure. For the whole process 1-2, the work done by the gas is given as :

$$W_{1-2} = \int_{V_1}^{V_2} p.dV$$

For a process 1-2, the gas expansion work is the area under the curve in the p-V diagram. It depends not only on the end state (initial and final) properties but also on the process between the states 1 and 2. For this reason, differentials like dp, dV etc. are called **'exact differential'** and a differential quantity such as p.dV is called an **'inexact differential'**. Similar to work, the heat interaction Q during a process also depends on the path (i.e. type of process). It is customary to denote exact differentials as dp, dV, dT etc. while the inexact differentials will be denoted using the symbol 'del' – for instance: δQ, δW etc.

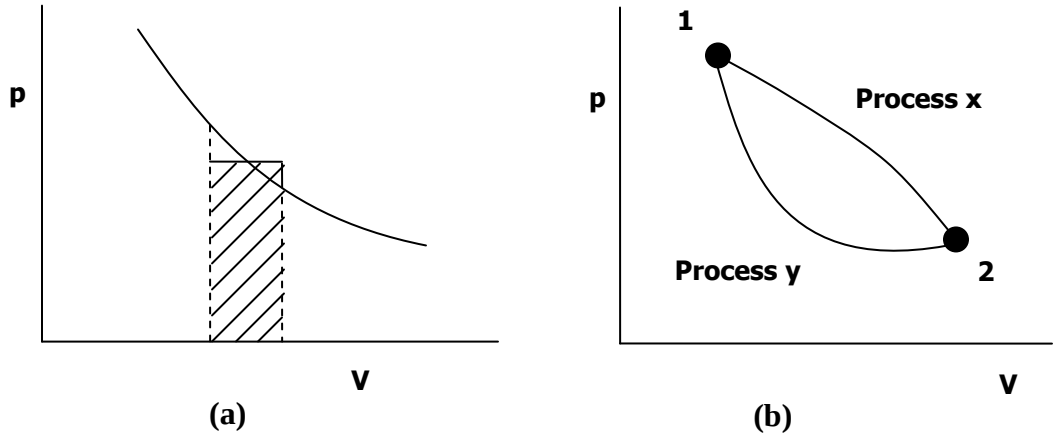


Fig. 1.4 State dependent and path dependent quantities

Pressure measurement :

We need to measure the absolute pressure in some applications and the relative pressure difference between two different locations in some cases. For example, for meteorological purposes, the absolute value of atmospheric pressure is measured on a daily basis and it is reported in the news papers each day. The device used for this purpose is the '**barometer**' which derives its name from Greek language (bar- pressure; meter- to measure). It consists of an inverted tube which is filled with mercury and kept immersed in a bath of mercury as shown in Fig. 1.5 a. This device gives the local atmospheric pressure as : $p_{atm} = \rho_{Hg}.g.h$ where g is the acceleration due to gravity, h is the height of mercury column read from the barometer and ρ_{Hg} is the density of mercury. The standard atmospheric pressure (1 atm) is defined as corresponding to a mercury column height of 760 mm and using values of g = 9.806 m/s^2 and density of mercury = 13600 kg/m^3, we get the standard atmospheric pressure as approximately equal to 101300 Pa or 101.3 kPa or 0.1013 MPa. Here 'Pascal' (denoted by the symbol Pa) is defined as equal to 1 Newton per square meter. In other words, 1 Pa = 1 N/m^2. Another convenient unit which is commonly used for pressure is the 'bar'. One bar is equal to 10^5 bar. Thus, 1 atm = 1.1013 bar.

A U-tube **manometer** is used to measure the pressure difference between two locations. Or, it could also be used to measure the pressure at a location, relative to that of the atmospheric pressure. The pressure at a point relative to that of atmosphere is known as the **'gauge pressure'** or **'vacuum pressure'**. For example, if the pressure at a point A is specified as 10 mm Hg gauge, it means that the absolute value of pressure at A corresponds to a mercury column height of (760 + 10 = 770 mm), assuming that the local atmospheric pressure is equal to the standard value of 760 mm Hg. Similarly, if the pressure at a is specified as 20 mm Hg vacuum, it implies that the absolute pressure at A corresponds to a mercury column height of 740 mm Hg. In general, a manometer may use any fluid (not necessarily mercury alone). The pressure difference between the two meniscii of the manometer is expressed as :

$$\Delta p = \rho.g.h$$

where ρ is the density of the manometric fluid, g is the acceleration due to gravity and h is the height difference between the two meniscii. Apart from the manometer, there are several types of pressure gauges. For instance, a device known as **'Bourdon gauge'** is commonly used to measure the pressure at some location in a pipe which transports a fluid. The Bourdon gauge consists of a soft metallic tube of elliptical cross-section, which deforms depending on the value

of local fluid pressure. The deformation of the tube is eventually calibrated in terms of the movement of a needle which denotes the corresponding gauge pressure ($p - p_{atm}$) in the fluid.

Chapter 2 WORK

In Mechanics, work is defined as the product of force into distance, given by the scalar product $W = \int \bar{F}.d\bar{s}$. Here 's' corresponds to the distance by which the point of application of the force moves. On the other hand, thermodynamic definition of work represents an energy interaction between the system and the surroundings. Work intrinsically is an inter-convertible form- in other words, it is always possible to convert work from one form to another with negligible losses. Suppose we do work in winding the spring of a wall- clock, almost all of that energy can be recovered for oscillating a pendulum (against solid friction and air friction). There are various forms of work: for instance, compression/ expansion of an elastic spring, gravitational work in lifting a mass, electrical work in charging a capacitor, compression/ expansion of a gas, magnetization of a piece of iron etc. are a few examples of work. Taking advantage of the inter-convertibility, in the modern day world, we transmit power to most house-holds, offices and factories as electricity, and locally convert it to any other desirable form using machines or appliances.

Keeping the above ideas in mind, the classical thermodynamic definition of work is given as follows:

"Work is done by a system during an interaction, if the sole effect external to the system could be reduced to the rise of a weight".

Here, the last phrase "rise of a weight", refers to the inter-convertibility aspect of work. Let us illustrate the above definition by way of a simple example.

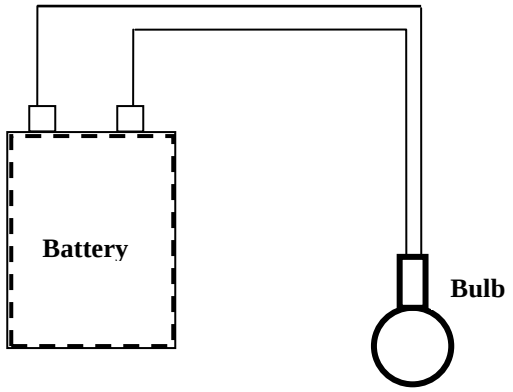


Fig. 2.1 Battery connected to an electrical bulb

Figure 2.1 illustrates the powering of an electric bulb by a battery. Let the voltage and current be 10 Volts and 5 Amps respectively. If the battery is operated for a period of 5 minutes, the energy consumed during the operation of the battery can be calculated as voltage x current x time = 10 V x 5 A x 300 s = 15000 Joules = 15 kJ. Considering battery as the system, we can consider this interaction as a positive work of 15 kJ done by the battery on its surroundings. How did we conclude that this is work interaction? Now let us remove the bulb and connect an electrical motor in its place. The electricity from the battery could have been supplied to the motor and the motor could have rotated a shaft (as is done in a ceiling fan). By tying a rope to the shaft which in turn is connected to a mass, we could have utilized the electrical energy for the lifting of a mass against gravity, as shown in Fig. 2.2.

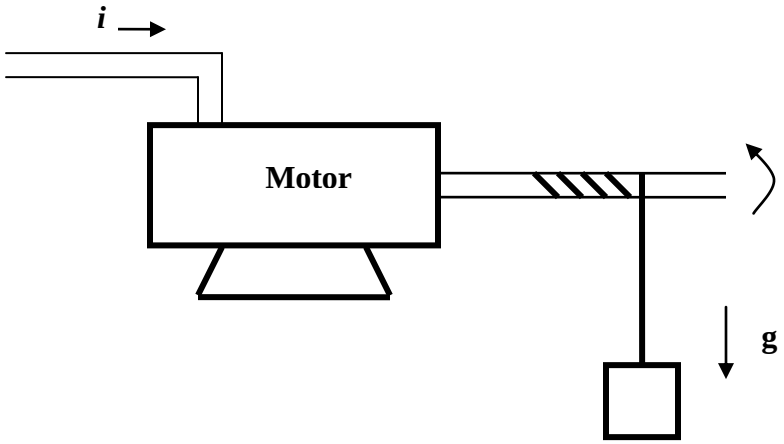


Fig. 2.2 Raising of weight by motor

The above example illustrates that although the real situation may involve the lighting of a bulb, the energy interaction from the battery is considered as work because the same energy could have been utilized for the rise of a weight. Another interesting feature is that in the lighting of the bulb, we can not identify any macroscopic movement of the point of action of a force by certain distance. So, the Mechanics definition of force x distance can not be applied here. (If you think that it may be possible to connect the movement of electrons through the wire to the amount of work done in this case, you will be in for a surprise. Whether the connecting wire from the battery to the bulb is 1 meter or two meters or 3 meters long, essentially the same amount of energy (~ 15 kJ) will be consumed in the process). Therefore, the energy based definition of Thermodynamics can provide interesting insights which are not possible with the Mechanics

definition of work. Due to inter-convertibility, all forms of work could be reduced eventually to the 'rise of a weight'. In other words, they could all be converted to gravitational work – the work associated with the rise of a mass.

The phrase "sole effect external to the system" in the definition of work has the following implication. When we replaced the bulb with a motor connected to a mass (through a rope tied to the motor shaft), the process undergone by the system (battery in this case) was left untouched. The same current of 5 A was drawn from the battery at the same ΔV of 10 V for the same duration of 5 minutes. The possibility of utilizing the associated energy of 15 kJ completely for the raising of a mass was looked into. If there are any frictional losses and electrical losses in the motor, they will result in heat dissipation; then, apart from lifting of the weight, there will also be slight changes in the temperature of the surrounding objects. Therefore, in order to satisfy the criterion of "sole effect", the motor should be an ideal one- with negligible losses. Usually, in most cases dealing with the conversion from one form of work to another, losses are indeed very small and almost 100% conversion is possible as you approach the ideal case. This is to be contrasted with the conversion from heat to work- where even with ideal processes, only about a third of the heat supplied gets converted into work.

Now let us look at other important aspects of work. The main concepts in this regard are listed below.

(i) Work is an <u>energy interaction</u> between the system and the surroundings. In other words, either system or surroundings should be the donor of energy- and the other should be the receiver.

(ii) When a system transfers energy in the form of work, it does positive work. At the same time, surroundings will do an equal amount of negative work. For instance, in the battery example, battery does positive work ie. $W_{battery} > 0$. The bulb does negative work ie $W_{bulb} < 0$.

(iii) In any problem, $W_{system} + W_{surroundings} = 0$ always. Work done by the system on the surroundings is defined as positive and work done by the surroundings on the system is taken as negative. That is why work is positive for the battery and negative for the bulb. The sign convention for work is based on the main interest to develop machines during industrial revolution, which would give work output by taking in heat input.

(iv) If 'n' different systems are having work interactions with each other, then $W_{sys1} + W_{sys2} + W_{sys3} + ... + W_{sysn} = 0$. Some of these systems may have positive work output and others may

have negative or zero. While considering any one system, all the other systems put together will form the surroundings for that system.

(v) A very important aspect of work interaction is that the process carried out by the system, must be resisted by the surroundings. If the surroundings do not offer any resistance, then the work done by the system will become zero.

Example 1: Free fall of a mass

Let us now consider the free fall of a particle of mass 'm' in vacuum, under the action of gravity. If the mass falls by a height 'h', according to the Mechanics definition, the work done during this process W = m.g.h. However, according to the Thermodynamic definition, work done by the mass W_{mass} = 0. In this case of free fall, although the force of body weight (= mg) moves the centre of gravity of the mass by a distance of h, actually no energy crosses the system boundary. As per the Thermodynamic definition, work is an energy interaction between the system (or CV) and the surroundings. Here, no energy interaction took place- only there was some internal energy conversion for the mass. The potential energy of the mass at higher elevation (initial position of the mass) was converted into its own kinetic energy as the mass accelerated during the fall. It is similar to a situation when a person takes out a 100 rupee note from one pocket, gets it converted to 10 notes of ten rupees and puts them in another pocket- in other words, there is money exchange in this transaction. In the case of the free falling mass, the total energy of the mass, i.e. potential energy + kinetic energy, remains the same. Since there is no energy interaction, work done has to be zero. An alternative way of understanding this example is to consider the fact that the surrounding vacuum does not offer any resistance to the motion of the mass; work exchange between the system and surroundings is possible only when the process is resisted by the surroundings.

Example 2: Fully resisted expansion of gas

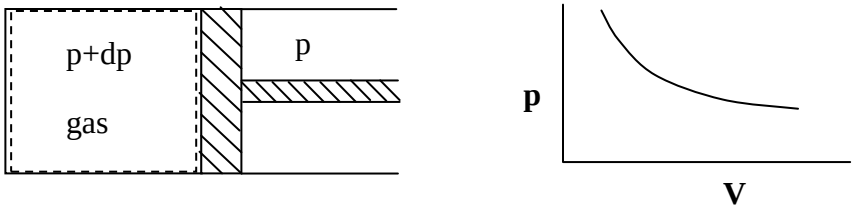


Fig. 2.3 Expansion of gas

Let us consider a gas which is enclosed inside a leak-proof frictionless piston-cylinder device. The gas pressure is slightly higher than the surrounding pressure and hence the piston starts moving slowly, causing slow expansion of the gas. Let the gas pressure at a particular instant be 'p+dp' and the corresponding external pressure exerted by the surrounding on the piston be 'p', as shown in Fig. 2.3. As the gas pressure varies during the expansion, the external pressure also varies such that the difference dp → 0. The gas expansion is fully resisted by the surroundings and hence, the piston moves slowly without any acceleration since the forces on the piston are in equilibrium. Such a slow gas expansion process is known as a 'fully resisted expansion process'. It can also be termed as a 'quasi- static process' because the system passes through a series of equilibrium states during the expansion.

Let us now contrast the above process with an 'unresisted expansion' situation, corresponding to the case when there is vacuum outside the piston. Here, the external pressure exerted by the surroundings is zero. If the piston which is initially held from movement by an agent, is suddenly released, will experience a very large acceleration. Both the piston and the enclosed gas which is adjacent to the piston start accelerating to very high velocities. Let us imagine a situation when the piston is very light and can be assumed to have negligible mass. In this case, the kinetic energy gained by the piston during its acceleration is negligible. Also, there is no matter present in the vacuum for energy exchange with the expanding gas. Thus, work done during this unresisted expansion case is zero. Note that acceleration of the gas also can not contribute to energy transfer since no system can do work on itself! Unless some energy transfer occurs between the system and surroundings, there is no question of any work done. During unresisted expansion, the pressure within the gas will not be uniform; the pressure near the close end will be high and the local pressure close to the piston will become nearly zero because of rapid acceleration. In such a case, the gas which is being considered as the system is not having internal equilibrium and it is not possible to denote properties such as system pressure, temperature etc. during the unresisted expansion. If eventually the piston comes to a stop and its properties become uniform (internal equilibrium is attained at the final state), then we can indicate the final state in the state diagram as the point 2. However, the intermediate states between the initial state 1 and the final state 2 can not be shown, because these intermediate states are not defined during the rapid gas expansion due to lack of equilibrium within the system. The zero work transfer in unresisted gas expansion can also be understood in an alternative way. A more precise definition for displacement work during the expansion/ compression of a fluid can be stated as: $W = \int_1^2 p_b \, dV_{sw}$

Where p_b is the boundary pressure and dV_{sw} is the differential swept volume for the moving boundary. In the case of unresisted expansion, the boundary pressure approaches a value of zero (because of vacuum on the outside). Thus, when the surroundings do not offer any resistance for the gas expansion process, it is not possible to have any work interaction.

Example 3: Resisted/ unresisted fall under the action of gravity

In order to illustrate the idea of fully-resisted, partially resisted and unresisted processes let us consider another example. In the event of an aircraft emergency, a passenger of 60 kg weight (~ 600 N force) decides to jump out of the aircraft and we are interested to find out the work done when the person descends by a height of 100 m. Let us consider three cases, namely: (a) when there is no air resistance (b) when air resistance does not fully meet the force due to the weight of the person (c) when the person ties himself onto a parachute and descends very slowly. In cases (a) and (b), since weight is not equal to air resistance, the person will accelerate. In case (c) when air resistance is equal to the weight, the person will descend slowly without any acceleration. As regards the amount of work done by the person, in case (a) work = zero, since no resistance is offered by the vacuum. In case (b), work done = air resistance x distance of fall = 50 N x 100 m = 5000 J = 5 kJ. In case (c), work done = air resistance x distance of fall = weight x distance of fall = 600 N x 100 m = 60000 J = 60 kJ.

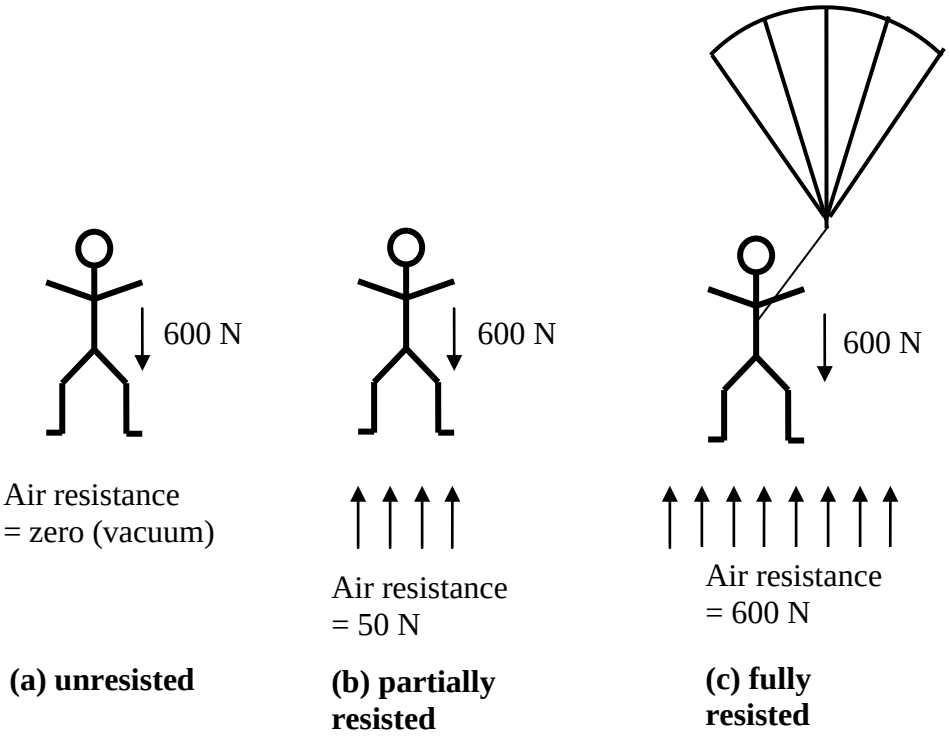


Fig. 2.4 Resisted/ unresisted fall

Displacement work for various processes :

The work done during the fully resisted expansion of a gas can be calculated if the variations of pressure and volume during the expansion process are completely known. Referring to Fig. 2.3, the force exerted by the gas on the piston is equal to (p.A) where the instantaneous gas pressure is p and A is the area of the piston (or cylinder). If the piston moves by a distance 'dx', then the amount of work done is equal to δW = p.A.dx = p.dV, where dV = A.dx = swept volume. The total work done during a process 1-2 can therefore be evaluated as:

$$W_{1-2} = \int_1^2 \delta W = \int_{V_1}^{V_2} p.dV$$

The work during a process 1-2 therefore is the area under the curve for the particular process in a p-V diagram. Let us now consider different processes in a p-V diagram.

(i) Constant pressure process (p = constant)

$$W_{1-2} = \int_{V_1}^{V_2} p.dV = p(V_2 - V_1) = p.\Delta V$$

(ii) A process with p.V = constant (Or, $p_1V_1 = p_2V_2 = C$).

$$W_{1-2} = \int_{V_1}^{V_2} p.dV = \int_{V_1}^{V_2} \frac{C}{V} dV = C \ln \frac{V_2}{V_1} = p_1 V_1 \ln \frac{V_2}{V_1} = p_2 V_2 \ln \frac{V_2}{V_1}$$

(iii) A process with $p.V^n$ = constant (Or, $p_1V_1^n = p_2V_2^n = C$).

$$W_{1-2} = \int_{V_1}^{V_2} p.dV = \int_{V_1}^{V_2} \frac{C}{V^n} dV = C(V_2^{1-n} - V_1^{1-n})/(1-n) = (p_2 V_2^n . V_2^{1-n} - p_1 V_1^n . V_1^{1-n})/(1-n)$$

$$Or, \quad W_{1-2} = \frac{(p_1 V_1 - p_2 V_2)}{n-1}$$

(iv) A constant process with V = constant.

In this case, since dV = 0, the work done is zero.

A constant pressure process is known as an *isobaric* process. A process with p.V = constant becomes a process with constant temperature, in the case of an ideal gas. The constant

temperature process is known as an *isothermal* process. The process pV^n = constant is known as a *polytropic* process and 'n' is called as the polytropic index. The case of n = γ (= ratio of specific heats, C_p/C_v) is called as an *adiabatic* process in the case of an ideal gas. This corresponds to the situation when the heat interaction with the gas is zero. In a special case, this situation also coincides with the case of constant entropy and therefore, it is also called as an *isentropic* process, as will be discussed later. Finally, the process of constant volume is known as an *isochoric* process. It is evident from the above discussions that the amount of work done depends on the process, apart from the end states 1 and 2.

Other forms of work:

(i) Spring work = $\frac{1}{2}k.x^2$ (for a linear spring) where k is the spring constant and x is the displacement.

(ii) Electrical work of a capacitor : $\frac{1}{2}C.V^2$ where C is the capacitance and V is the voltage.

(iii) Electrical work for a resistor : $i^2R\ x\ t$ where i is the current through the resistor, R is the resistance and t is the time of operation.

(iv) Shaft work : $\frac{2\pi NT}{60}\ x\ t$ where N is the shaft speed (in revolutions per minute) and T is the torque. Also t is the time of operation.

(v) Gravitational work : m.g.h where m is the mass, g is the acceleration due to gravity and h is the height by which the mass is raised or lowered.

(vi) Surface tension work: σ x ΔA, where σ is the surface tension and ΔA is the increase in area of the film.

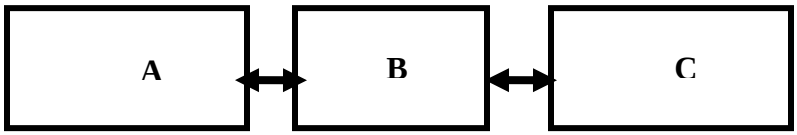
In all the examples discussed above, the expression may be used to calculate only the magnitude of the work. The actual assignment of positive or negative sign for the work may be given depending on whether the system gives out or receives energy in the form of work. For instance, in the case of a linear spring, work is negative whether the spring is extended or compressed. Spring work can be positive only when an already extended (or compressed) spring return back to its initial state by releasing the stored energy.

Chapter 3 Temperature and Heat

In Chapter 1, temperature was described as an intensive property of a system. In common parlance, we understand temperature as a property which is related to the degree of hotness (or coldness) of a substance for physical touch. It is to be noted however, that human perception regarding the degree of hotness or coldness of an object could be deceptive- because what our body senses is only the loss or gain of heat from the surroundings. For example, on a cold winter night, the atmosphere may appear to be colder when you are moving on a vehicle than when you are stationary. This is because of increased heat loss from the body to the air while in motion (due to convective effects). Similarly, on a cold morning, a metallic object appears to be colder than a wooden object, although both are at the same temperature. This can also be attributed to larger heat loss from the body to the metal due to its higher thermal conductivity. Such subjective assessments can be avoided if we employ calibrated devices for measuring temperature.

The measurement of temperature hinges on the idea of equilibrium between systems. System **'A'** is said to be in equilibrium with another system **'B'**, if no changes are observed in the properties of both the systems, when they are in contact with each other. For example, if both systems are gases and pressure of **A** is the same as pressure of **B**, then the two systems are said to be in mechanical equilibrium (i.e. forces are in balance). If the two pressures are not equal, then a work interaction process will occur between the two systems until the pressures become equal. Similarly, if the temperature of **A** and temperature of **B** are equal, then the two systems are said to be in thermal equilibrium. If the temperatures of the two systems are not equal, then a heat interaction process will occur, until thermal equilibrium occurs between the systems. As a consequence of heat flow, changes in some properties can be observed for these systems and the observed property changes can be calibrated to indicate the temperature changes.

The **Zeroth Law of Thermodynamics**, as stated below, further illustrates the idea of equilibrium: **" If system A is in equilibrium with system B and system B is in equilibrium with system C, then systems A and C must be in equilibrium with each other."**



Here, system **B** could be a thermometer, for instance. If **B** shows a temperature reading of 100 units while in equilibrium with **A**, and it shows the same temperature reading of 100 units while in equilibrium with system **C**, then systems **A** and **C** must both be at the same temperature. Attaining thermal equilibrium is essential for obtaining the correct reading. How many times have we felt irritated when the doctor leaves the thermometer in the mouth for a couple of minutes, in order to read the body temperature correctly when we have fever? If the thermometer is taken off soon, it would not have reached thermal equilibrium with the body and the temperature value read will be incorrect.

Now let us turn our attention to the methods for measuring temperature. Any property of a substance that varies with temperature could be used for measuring temperature, after proper calibration of the associated device. For example, in a mercury thermometer, the length of a mercury column varies with temperature due to thermal expansion. Expressing the column length as a suitable function of temperature, the device can be calibrated by measuring the length at some known values of temperature. For instance, the variation of column length (ℓ) could be written as a linear function of temperature in the form: $\ell = a + b.T$, if the temperature variation is not too large. For the purpose of calibration, we could use the known temperatures such as the ice point 0°C and the steam point 100°C. Here, 'ice point' refers to the melting point of ice at 1 atm pressure and 'steam point' refers to the boiling point of pure water at 1 atm pressure. The constants 'a' and 'b' can be calculated as:

$$a = \ell_0 \quad and \quad b = \frac{\ell_{100} - \ell_0}{100}$$

in terms of the mercury column lengths corresponding to the steam and ice point temperatures. A similar calibration procedure could be adopted for any other type of thermometer also. Liquids other than mercury may be used in a liquid-in-glass thermometer. Also, high temperature materials such as quartz may be employed instead of ordinary glass. Such modifications can change the operational temperature range of the thermometer.

Temperature can also be measured by employing principles other than the linear thermal expansion of a substance. For example, the ideal gas behavior which relates the gas pressure, temperature and volume can be used to measure temperature. A thermometer constructed on this principle will be called as an 'ideal gas thermometer'. Such a thermometer could be used in two modes, namely: the 'constant volume mode' in which volume is constant and the pressure varies linearly with temperature; alternatively, it can be used in the 'constant pressure mode' for which the volume varies linearly with temperature. For the constant volume mode, expressing

the gas pressure as p = a + b.T, the constants 'a' and 'b' can be obtained as discussed earlier. Indeed, for a gas of mass m, such a calibration will result in the expression

$$pV = m.\left(\frac{\bar{R}}{M}\right)(T + 273.15)$$

for the relationship between the gas pressure, volume and temperature. Here, T is expressed in degree Celsius. It is also possible to define a new temperature scale known as the Kelvin scale, where temperature in Kelvin is given as 'temperature in degree Celsius + 273.15'. Note that for T = - 273.15^{o}C, p → 0. In other words, the calibration of the ideal gas thermometer should give:

$$\frac{p_{steam\ pt} - p_{ice\ pt}}{100 \times p_{ice\ pt}} = \frac{b}{a} = \frac{1}{273.15}$$

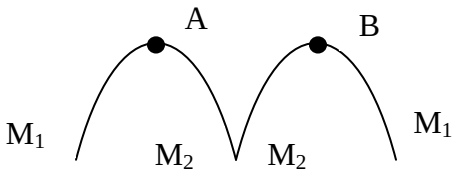
From a microscopic point of view, temperature can be related to the average kinetic energy of the randomly moving molecules of a gas. For instance, in monatomic gases such as helium, argon, neon, etc. the average kinetic energy in each of the three directions (x,y,z) is given by:

$$\frac{1}{2}m\bar{u}^2 = \frac{1}{2}m\bar{v}^2 = \frac{1}{2}m\bar{w}^2 = \frac{1}{2}\kappa T$$

Where κ is the Boltzmann's constant and $\bar{u}, \bar{v}\ and\ \bar{w}$ are the average molecular velocity components for the three directions under equilibrium conditions. Here, T is the temperature expressed in Kelvin. It is clear that as T approaches zero Kelvin, all molecular motion will cease and the pressure also will approach zero value. The pressure can be related to the molecular momentum, from the microscopic point of view.

The other commonly employed techniques for measuring temperature are described below:

(i) Thermo-electric effect: Let A and B represent two junctions made of two different metals M_1 and M_2 as shown in the figure. If the temperatures of A and B are different, then an emf (voltage) will develop. By measuring this emf, temperature can be measured. In practice, one of the junctions (cold junction) will be kept at a known temperature such as ice point and the hot junction temperature could be evaluated from the measured value of voltage. The voltage could be expressed as a function of the hot junction temperature (T) such as emf = a + b.T + c.T^2, where the constants



a, b and c could be evaluated from calibration. These types of hot and cold junctions are known as 'Thermocouples'.

(ii) Dependence of resistance on temperature: In devices known as resistance thermometers, the electrical resistance of a wire may vary with temperature, typically in a polynomial form. For instance, if $R(T) = a + b.T + c.T^2 + d.T^3$, by measuring the resistance of the wire at four known temperatures, it is possible to determine the constants a, b, c and d. Obviously, such a procedure involves the calibration of the resistance thermometer. There are also some semiconductor based devices known as thermistors for which the electrical resistance does not vary as a polynomial- but it varies exponentially with respect to temperature. The associated constants can be determined through calibration for thermistors also.

(iii) Liquid crystals: For liquid crystals, the color of the crystal depends on its temperature. By relating color and temperature, it is possible to use the liquid crystal as a thermometer.

(iv) Pyrometers: The temperature of a body which is at an elevated temperature can be measured with the help of the radiation emitted by the body. Devices which measure temperature from the radiation emitted by a body are known as pyrometers.

For calibrating different types of thermometers, a list of known temperatures is necessary. Apart from the ice point and steam point, several other temperature values are required. These are typically phase change temperatures such as the boiling points of liquids, freezing points of metals, etc and they are called as 'Fixed Points'. A few fixed points which are commonly employed are: Oxygen point (boiling point of liquid oxygen = -183.0^{o}C), Sulfur point (boiling point of liquid sulfur = 444.7^{o}C), Antimony point (freezing point of antimony = 630.7^{o}C), Silver point (freezing point of liquid silver = 961.9^{o}C) and Gold point (freezing point of liquid gold = 1064.4^{o}C). Depending on the number of constants to be determined, an equal number of fixed point temperatures can be selected and employed for calibrating the temperature measuring device. Any calibrated device will indicate accurate temperature values at the fixed temperatures used for the calibration. However, at other temperatures, the measured values may have some intrinsic error. For example, consider two different devices- say, a mercury thermometer which is calibrated using the ice point, steam point and a linear interpolation function; a thermocouple which is calibrated using the temperatures of ice point, steam point, sulfur point and a quadratic interpolation function. Although both devices will read ice point as 0°C and steam point as 100°C, it is possible that the mercury thermometer may read a temperature value as 50°C while the thermocouple may report the same temperature as 50.5^{o}C. Such small differences will exist between the data measured by different thermometers, at all temperatures except the fixed points used for calibration.

In Chapter1, heat was described as an energy interaction between the system (or CV) and the surroundings. A more precise definition for heat is that **it is an energy interaction which occurs because of the temperature difference between the system (or CV) and the surroundings**. Heat transfer between two objects may occur by heat conduction, convection or radiation. Conduction heat transfer occurs because of molecular collisions between two objects which are in physical contact with each other. Consider two systems A and B, with the temperature of A being greater than the temperature of B (i.e. $T_A > T_B$). If the two systems are in physical contact, at the interface, molecules of A will randomly collide with those of B. Since $T_A > T_B$, the average molecular velocity of system A will be larger than the average molecular velocity of B. In the process, some random kinetic energy of molecules of A will be transferred to those of B due to collisions. Such energy transfer is known as heat transfer. Convection heat transfer occurs when there is relative motion between a fluid and an object. As the relative velocity increases, the rate of convective heat transfer also increases. Radiative heat exchange occurs through space. Even when two objects at different temperatures are not in physical contact, photons emitted by one can be received by the other through space. When an object is at a high temperature, some of its molecules can attain a higher energy state as a consequence of the energy exchange due to molecular collisions. When these higher energy state molecules return their ground states, they emit photons corresponding to the energy difference. Such photon exchange between hot and cold objects gives rise to radiative heat transfer. The rate of heat transfer by the three modes can be summarized as follows:

$$Q_{conduction} = -kA\frac{dT}{dx};\ Q_{convection} = hA(T_A - T_B)\ and\ Q_{radiation} = \sigma\varepsilon A(T_A^4 - T_B^4)$$

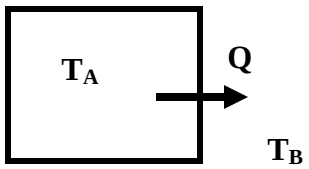
The radiative heat transfer expression corresponds to the heat exchange between an object at temperature T_A and the surrounding at temperature T_B. In the above relations, k is the thermal conductivity of the medium, A is the surface area, h is the heat transfer coefficient, σ is the Stefan- Boltzmann constant and ε is the emissivity of the surface.

Some of the popular misconceptions about heat are listed below:

(i) Whenever an object receives heat, its temperature will increase – **Wrong**. During a phase change process such as boiling, melting etc. even though there may be heat addition, the temperature will not increase.

(ii) If the temperature of an object increases during a process, there must have been a heat input to that object – **Wrong**. Temperature of an object can be increased by work input also. In smithy shop, as mechanical working is carried out on an object, its temperature increases.

In the context of these clarifications, revisit the above definition given for heat interaction.



In the figure shown above, when system temperature T_A is greater than the surrounding temperature T_B, heat is transferred from the system to the surroundings. The sign convention adopted is as follows: heat added to a system is positive and heat removed from a system is negative. In other words, when heat transfer occurs from the surroundings to the system, $Q > 0$. If the heat transfer is from the system to the surroundings, then $Q < 0$. Both heat and work are energy interactions occurring across the system (or CV) boundary. Depending on how the system boundary is drawn, the classification of the interaction as heat or work could change. For example, consider a vessel filled with water which is heated by an electrical coil wound on its outside. If the system boundary is drawn such that it includes the electrical coil, then the electrical energy flowing through the coil will be treated as work. On the other hand, if the system boundary excludes the coil (and includes only the vessel with water), then the energy interaction occurring between the vessel and the coil is heat. In fact, this energy interaction takes place because the coil gets red hot due to current flow and heat is transferred from the coil to the vessel by virtue of the temperature difference.

Chapter 4 First Law of Thermodynamics for a System

Although heat and work interactions were known independently for a long time, a detailed understanding of their inter-relationship was not known until the mid-19th century. Also, it was not realized that heat is a form of energy interaction. The prevailing theory for heat in the early times was the calorific theory, which likened heat to an invisible fluid. It was thought that just as water flows from a higher level to lower level, heat flows from a higher temperature to a lower temperature. It was James Prescott Joule, who between the years 1840 to 1849, clearly showed the equivalence of heat and work through his experiments. He computed mechanical work W as the product of force into distance, and heat in terms of the temperature rise using the expression $Q = m.c.\Delta T$. He demonstrated that the magnitudes of work input and heat input required for achieving the same temperature rise in a substance always had a fixed ratio, $W/Q = 4.186$ N.m/Calorie, which is known as the "Mechanical equivalent of heat". The set up used by Joule for arriving at this conclusion is shown in Fig. 4.1. It consists of a known mass of water whose temperature is raised by the operation of a paddle wheel. The work input for paddle wheel is calculated as torque x angular displacement; the torque in turn, is given as the force due to the mass hung from the rope x radius of the shaft which drives the paddle wheel. Equivalent heat is calculated as Q = mass of water x specific heat of water x ΔT for water.

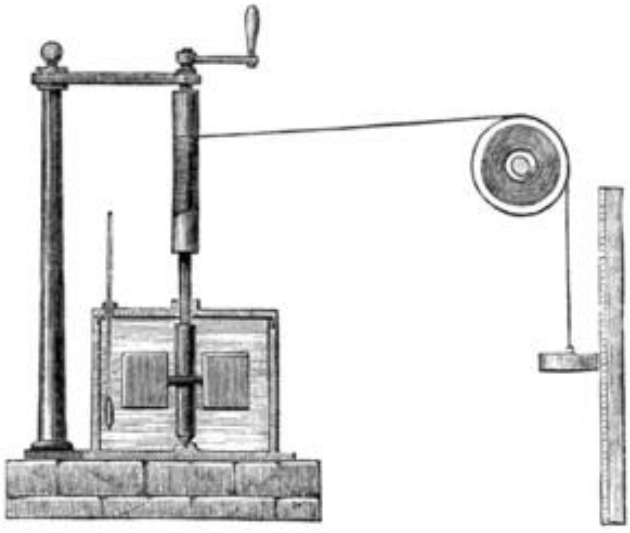


Fig. 4.1 Joule's experiment for finding mechanical equivalent of heat

The above- mentioned experiment of Joule conclusively proves that both heat and work are energy interactions. It also provides another interesting insight about cyclic processes. Let us denote the work needed to raise the temperature of water by ΔT as W. Now the heat (in Calories) which has to be removed from water to bring it back to the initial temperature is equal to W/$\mathcal{J}$ where $\mathcal{J}$ is the mechanical equivalent of heat (= 4.186 N.m/Calorie). Thus, irrespective of how long the paddle wheel is operated or what is the ΔT achieved, the heat to be removed for restoring the water back to its initial temperature Q = W/$\mathcal{J}$. This result can be generalized further for any cycle with many work and heat interactions as:

$$\sum_{cycle} W = J \sum_{cycle} Q \qquad (4.1)$$

In other words, for a cyclic process, the net heat and net work interactions during the cycle are always related through Eq. (4.1). This form of equation is preferable when there is a finite number of sub-processes in the cycle (see, Fig. 4.2 a). In the case of continuous changes without any identifiable sub-processes for the cycle (see Fig. 4.2 b), the more appropriate form is:

$$\oint \delta W = J \oint \delta Q \qquad (4.2)$$

In both equations (4.1) and (4.2), the constant $\mathcal{J}$ performs the job of converting from the heat unit to work unit. In other words, if the same units are employed for heat and work (say kJ), then the conversion constant $\mathcal{J}$ = 1. In such a case, equations (4.1) and (4.2) can be rewritten as:

$$\sum_{cycle} W = \sum_{cycle} Q \qquad (4.3)$$

and,

$$\oint \delta W = \oint \delta Q \qquad (4.4)$$

The above equations (4.1) to (4.4), are said to be the forms of I Law of Thermodynamics, which are applicable to a system (of fixed mass) undergoing a cyclic process.

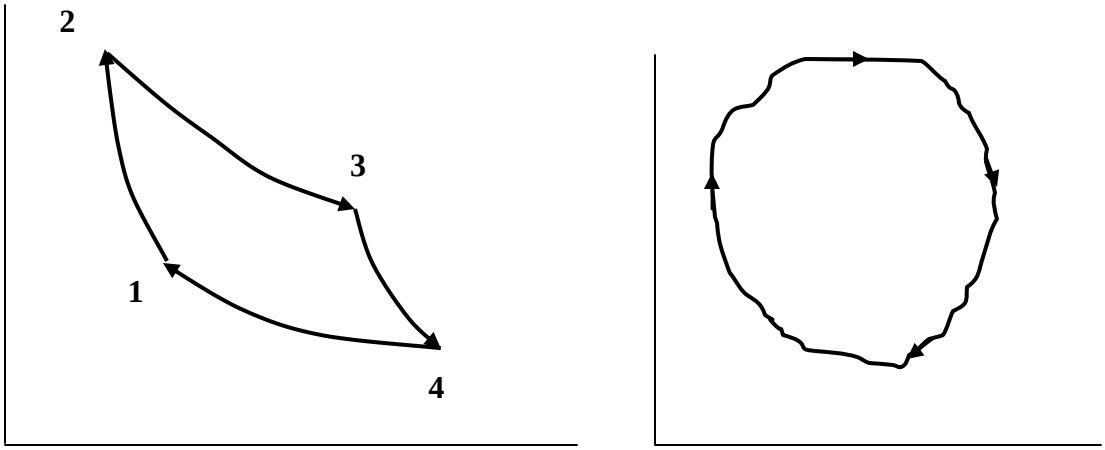


Fig. 4.2 (a) Cycle with sub-processes (b) cycle without any sub-processes

For the cycle shown in Fig. 4.2(a) with four sub-processes 1-2, 2-3, 3-4 and 4-1, Eq. (4.3) can be written as:

$$W_{1-2} + W_{2-3} + W_{3-4} + W_{4-1} = Q_{1-2} + Q_{2-3} + Q_{3-4} + Q_{4-1}$$

The first law for a cyclic process can be understood as follows: Since the system returns to its initial state at the end of the cycle, there is no net storage of energy in the system. Therefore, the net heat and work interactions between the system and surroundings should balance (i.e. input = output) for the whole cycle, irrespective of whether their values are positive or negative (Fig. 4.3).

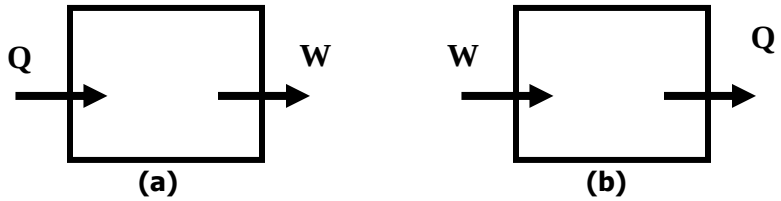


Fig. 4.3 Energy Interactions

For a non- cyclic process, the heat and work interactions need not be equal. In that case, the difference between the two will be the stored energy during the process. Considering positive heat input to the system and positive work output by the system (Fig. 4.3a), the I law can be expressed as

$$\mathbf{Q - W = \Delta E.} \qquad (4.5)$$

For instance, if heat input is 10 kJ and work output is 6 kJ, the change in total stored energy ΔE is 4 kJ. Now, it is easy to see that Eq. (4.5) is valid even if Q and W are negative. Referring to Fig. (4.3b), if W = - 10kJ and Q = - 6 kJ, the stored energy is still ΔE = 4 kJ as per Eq. (4.5). In the cases where input energy interaction is less than the output, ΔE of the process can be negative.

In Eq. (4.5), Q and W are path dependent quantities, while the total stored energy E is a property. Change in E depends only on the end states and not on the path followed in the process. This can be illustrated using the first law for a cyclic process. In order to prove that E is a property, consider the two cycles 1-A-2-C-1 and 1-B-2-C-1 as shown in Fig. 4.4.

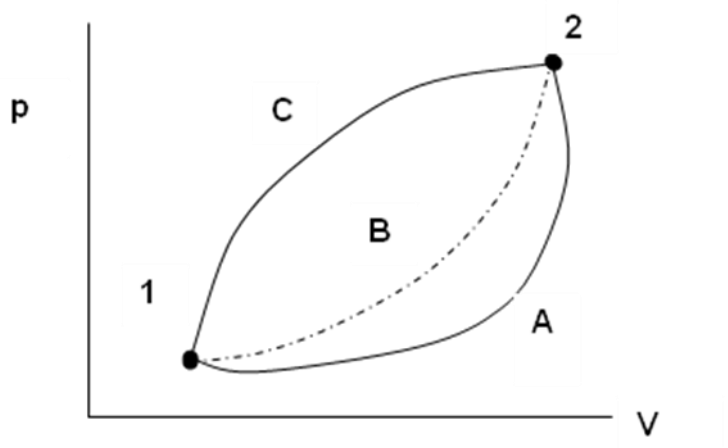


Fig. 4.4 Proof for E as a property

For the cycle 1-A-2-C-1, application of I law gives:

$$Q_{1-A-2} + Q_{2-C-1} = W_{1-A-2} + W_{2-C-1} \qquad (4.6)$$

Similarly, for the cycle 1-B-2-C-1, we get:

$$Q_{1-B-2} + Q_{2-C-1} = W_{1-B-2} + W_{2-C-1} \qquad (4.7)$$

Subtracting Eq. (4.7) from Eq. (4.6) gives:

$$Q_{1-A-2} - Q_{1-B-2} = W_{1-A-2} - W_{1-B-2} \qquad (4.8)$$

The above equation can be rearranged as:

$$Q_{1-A-2} - W_{1-A-2} = Q_{1-B-2} - W_{1-B-2} \qquad (4.9)$$

Since no assumptions have been made about the processes 1-A-2 and 1-B-2, it is evident that for two given end states 1 and 2, the difference between Q_{1-2} and W_{1-2} is path independent.

Having shown that the total stored energy E is a state function (property), let us now understand its intrinsic characteristics. The total energy E may be the sum total of different forms of energy, namely: kinetic energy, potential energy, electrical energy, magnetic energy, internal energy, etc. Each of these energy forms, in turn, may be linked with a particular property such as the velocity, elevation etc., for a system of given mass m. Thus, change in the total stored energy E may be expressed as:

$$\Delta E = \Delta KE + \Delta PE + \Delta EE + \Delta ME + \Delta U + ... \qquad (4.10)$$

The properties associated with the individual energy forms (at fixed mass) are: KE- velocity; PE- elevation; EE- charge; ME- magnetization; U – Temperature. For a system which does not have energy storage in the form of kinetic energy, potential energy, electrical energy, etc. the first law (Eq. 4.5) can be simplified to the form:

$$Q - W = \Delta U \qquad (4.11)$$

Consider for example, a gas undergoing expansion in a piston- cylinder device as a consequence of heat addition (Fig. 4.5). If the heat added is Q = 10 kJ and the work done by the gas is W = 6 kJ, then the increase in the internal energy of the gas will be ΔU = 4 kJ. The increase in the internal energy of the gas will reflect as an increase in the temperature of the gas.

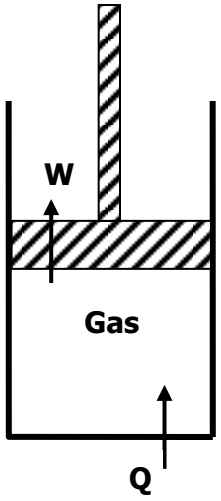


Fig. 4.5 Expansion of a gas due to heat addition

For an infinitesimal change, equation (4.11) can be rewritten as:

$$\delta Q - \delta W = dU$$

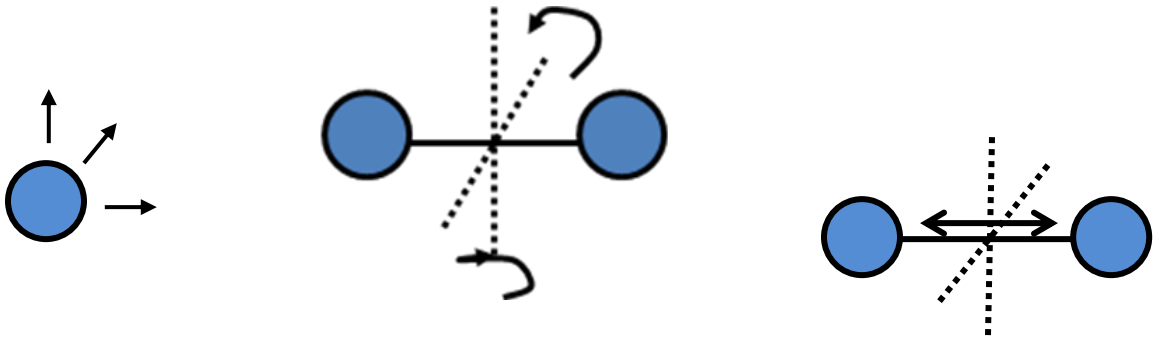
Substituting for the expansion work of the gas as p.dV, we get:

$$\delta Q = pdV + dU \qquad (4.12)$$

Note that in the above expressions, heat and work are represented as δQ and δW, while the internal energy change is written as dU, in order to differentiate the exact and inexact differentials.

Internal energy is the energy associated with the random motion of molecules (and bond energies between the atoms which form the molecules). The molecular motion, in turn, can involve the translation of molecules in 3-D space and therefore, at any instant, one can describe the motion in terms of the x-,y- and z- velocity components of each molecule. The velocity components keep changing with respect to time because of collisions between molecules. For substances having more than one atom per molecule (eg. N_2, H_2, O_2, CO_2, H_2O), apart from translational motion, there could also be rotation or vibration. Detailed analysis based on kinetic theory shows that the energy associated with each mode of random molecular motion (under equilibrium conditions) is equal to ½ κT, where κ is known as the Boltzmann's constant and T is the

temperature in Kelvin. Thus, if one considers the nitrogen molecule which could exhibit three modes of translation, two modes of rotation and one mode of vibration under certain conditions (see Fig. 4.5), the associated kinetic energy per molecule will be 6 x ½ κT. Adding up the energy of random motion (and the bond energies which are essentially constants) associated with all the molecules in a system, gives rise to the internal energy of the system. It is evident from the above discussion that the internal energy U of a system, primarily depends on its temperature.



(a) Translation **(b) Rotation** **(c) Vibration**

Fig. 4.6 Molecular motion

The internal energy of a system can be expressed in a form U = U(T). Since the internal energy of the whole system is an extensive property which depends on the mass of the system, it is often convenient to deal with internal energy per unit mass of the system. This is known as the specific internal energy and is denoted by the symbol 'u'. Considering the I law form of Eq. (4.12), it is clear that for a constant volume process (dV = 0), the heat transfer Q = ΔU. Now, the specific heat at constant volume which is defined as the amount of heat required to raise the temperature of unit mass of the substance by 1°C while maintaining the volume constant, can be expressed as:

$$C_v = \left(\frac{\Delta U}{m \times \Delta T} \right)_{v=const} = \left(\frac{\partial u}{\partial T} \right)_v \qquad (4.13)$$

As can be seen from equation (4.12), the units for internal energy and specific internal energy are J (Joule) and J/kg respectively. The unit of C_v is J/(kg K).

Now let us define another property called Enthalpy (H), which is useful in constant pressure processes of a system or the flow processes of a control volume, as given below:

$$H = U + pV \qquad (4.14)$$

The second term on the right hand side of equation (4.14) is known as flow work. It is associated with the work done by a fluid element on the fluid ahead of it, during a flow process. The change in enthalpy can written as

$$dH = dU + pdV + Vdp \qquad (4.15)$$

For a constant pressure process, dp = 0 and hence,

$$dH = dU + pdV = \delta Q \qquad (4.16)$$

as per Eq. (4.12). It is evident from Eq. (4.16) that for a constant pressure process, the heat transfer is given by

$$Q_{1\text{-}2} = \Delta H = H_2 - H_1 \qquad (4.17)$$

Now the specific heat at constant pressure which is defined as the amount of heat required to raise the temperature of unit mass of the substance by 1°C while maintaining the pressure constant, can be expressed as:

$$C_p = \left(\frac{\Delta H}{m \times \Delta T} \right)_{p=const} = \left(\frac{\partial h}{\partial T} \right)_p \qquad (4.18)$$

In the above expression, 'h' is the specific enthalpy which is enthalpy per unit mass. The units for enthalpy, specific enthalpy and specific heat at constant pressure are given by J, J/kg and J/(kg K) respectively. The specific heats can be viewed as the slopes of the internal energy variation or enthalpy variation curves with respect to temperature along V = constant or p = constant paths respectively, as shown in Fig. 4.7. In this sense, the specific heats are also properties, which are point (state) functions.

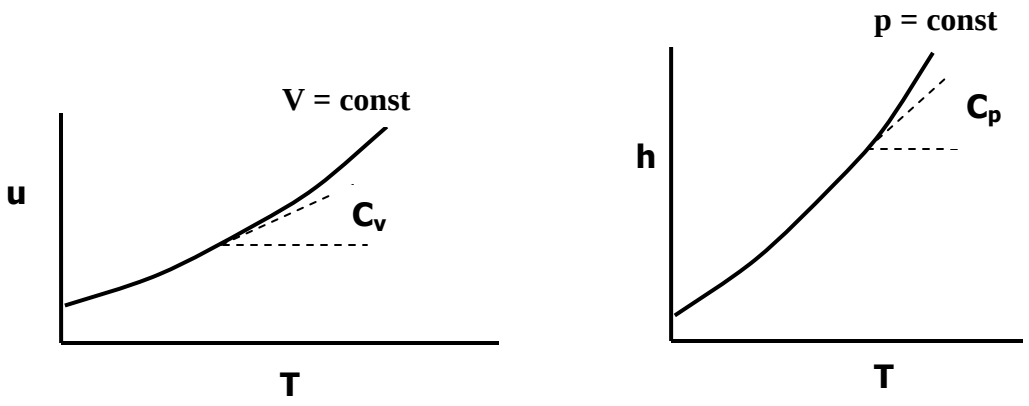


Fig. 4.7 Specific heats at constant volume and constant pressure

Chapter 5 Properties of a Pure Substance

5.1 Definition of Pure Substance

The state of a system is defined in terms of its properties. The number of properties needed to define the state uniquely depends on the type of substance which constitutes the system. For example, in the case of a chemically reacting system, to define the state of the substance at any instant, one may need to prescribe the concentration of reactants and products, in addition to the pressure and temperature. The state of a rigid solid body moving under the action of gravity (and having no electrical/ magnetic effects) could be defined in terms of its temperature, velocity and elevation from a datum.

We define **'pure substance'** as a **substance which has uniform composition and chemical aggregation** throughout the process under consideration. For example, air is a mixture primarily made up of nitrogen and oxygen, with small fractions of CO_2, water vapor and inert gases. In problems involving compression/ expansion of air at moderate pressures and temperatures, it could be treated as a pure substance. However, in a process involving liquefaction of air, we cannot treat air as a pure substance because of difference in the boiling points of nitrogen and oxygen; in fact, the compositions of the liquid and the vapor phases may not be the same and the liquid may contain higher concentration of oxygen than the vapor. To cite another example, during the combustion of methane- air mixture, the mixture cannot be treated as a pure substance. This is because of the fact that the mixture composition varies with time during the combustion process. On the other hand, the mixture of steam and water enclosed in a vessel can be treated as a pure substance- as per the definition given above.

For a pure substance, the **two-property** rule is applicable as described below:

"The state of a pure substance of given mass can be specified in terms of two independent properties, in the absence of effects due to gravity, motion, electricity, magnetism, elastic deformation, etc."

In other words, if the substance has only one mode of energy storage –namely, that of internal energy, two independent properties are adequate to define its state (for given mass). For example, consider a gas of mass m which is enclosed inside a cylinder with the help of a piston. The initial state can be specified in terms of the initial pressure and volume, p_1 and V_1 respectively. Similarly the final state may be given in terms of the final pressure and volume, p_2 and V_2 respectively. Such specification of the states of the gas implies that all the other properties of state 1 or state 2 can be determined from the given pressure and volume data. In other words, there are certain relationships between the different properties of a substance. In this chapter, we elaborate on the relationships between various properties for different types of pure substances. The substances that will be considered are: (i) ideal gas (ii) mixture of ideal gases (iii) steam/ water substance.

5.2 Properties of ideal gas

Any gas can be treated as an ideal gas if its molecules exert negligible attractive (van der Waal) forces on each other. Since this condition becomes applicable for molecules which are separated wide apart, the ideal gas assumption becomes valid as $V \rightarrow \infty$ and $p \rightarrow 0$. An ideal gas obeys the Boyle's and Charles' laws which can be stated as follows:

Boyle's law: "At constant temperature, the volume of a given mass of gas is inversely proportional to its pressure".

Charles' law: "At constant pressure, the volume of a given mass of gas is directly proportional to its temperature"

Combining both the laws, the p-V-T relationship for an ideal gas can be expressed as

$$\frac{pV}{T} = const\,.$$

More elaborate expressions involving the amount of gas can be written as:

$$pV = n\bar{R}T \qquad (5.1)$$

Or, $$pV = mRT \qquad (5.2)$$

In the above equations, n – number of moles (or kmols) of gas; T – temperature in Kelvin; $\overline{R}$ - Universal gas constant or molar gas constant = 8314 J/kmol K; m – mass of gas in kg; R – specific gas constant = $\frac{\overline{R}}{M}$ J/kg K where M is the molecular weight of the gas (kg/kmol). Also, p is the pressure (Pa) and V is the gas volume (m^3). The quantity of gas can be specified in terms of its mass m (kg) or as kmol, where kmol is defined as molecular weight expressed in kg. For instance, one kmol of nitrogen gas corresponds to 28 kg, while one kmol of hydrogen would be having a mass of 2 kg.

Equations (5.1) and (5.2) can be written in many different forms. For instance,

$$p\frac{V}{n} = p\overline{V} = \overline{R}T \qquad (5.3)$$

Where $\overline{V}$ is the molar volume (m^3/kmol). Also,

$$p\frac{V}{m} = RT \rightarrow pv = RT \qquad (5.4)$$

$$\frac{p}{\rho} = RT \rightarrow p = \rho RT \qquad (5.5)$$

In the above expressions, v denotes the specific volume (m^3/kg) and density ρ = 1/v (in kg/m^3).

The main criteria to be satisfied by an ideal gas of given mass are:

(i) Pressure x Volume/ Temperature = constant.
(ii) Internal energy, U = U(T) only.
(iii) Enthalpy, H = H(T) only.

In chapter 4, it was discussed that the internal energy per degree of freedom (translational, rotational or vibrational motion of a molecule) in a gas is equal to ½ κT

under equilibrium conditions. Here, the Boltzmann constant κ is equal to $\frac{\overline{R}}{N_{AV}}$, where N_{AV} is the Avagadro number, or the number of molecules per mole (or kmol). The value of N_{AV} = 6.023 x 10^{26} molecules/ kmol. If the number of translational, rotational and vibrational degrees of freedom per molecule is equal to 'D', then the internal energy per molecule = D x ½ κT. For one kilomole (kmol) of the gas, the molar internal energy $(\overline{U})$ is given as:

$$\overline{U} = N_{AV} \times D \times \frac{\kappa T}{2} = D \times \frac{\overline{R}\,T}{2} \tag{5.6}$$

The molar enthalpy is given as:

$$\overline{H} = \overline{U} + p\overline{V} = \overline{U} + \overline{R}T = \frac{D+2}{2}\overline{R}T \tag{5.7}$$

The molar internal energy and molar enthalpy can also be expressed in terms of the molar specific heats as

$$\overline{U} = \overline{C}_v T = D \times \frac{\overline{R}\,T}{2} \; where \; \overline{C}_v = D \times \frac{\overline{R}}{2} \tag{5.8}$$

Similarly, $\overline{H} = \overline{C}_p T = (D+2) \times \frac{\overline{R}\,T}{2} \; where \; \overline{C}_p = (D+2) \times \frac{\overline{R}}{2}$ (5.9)

The molar internal energy and molar enthalpy are expressed in the units of J/kmol, while the molar specific heats are expressed in the units of J/kmol K. On a per unit mass basis, the specific internal energy and specific enthalpy (in units of J/kg) can be written as

$$u = \frac{\overline{U}}{M} = D \times \frac{\overline{R}\,T}{M \times 2} = D \times \frac{RT}{2} = C_v T \; where \; C_v = D \times \frac{R}{2} \tag{5.10}$$

$$h = \frac{\overline{H}}{M} = (D+2) \times \frac{\overline{R}\,T}{M \times 2} = (D+2) \times \frac{RT}{2} = C_p T \; where \; C_p = (D+2) \times \frac{R}{2} \tag{5.11}$$

The specific heat ratio (γ) is given as

$$\gamma = \frac{\overline{C}_p}{\overline{C}_v} = \frac{C_p}{C_v} = \frac{D+2}{D} \tag{5.12}$$

in terms of the number of degrees of freedom. As discussed in chapter 4, for a diatomic gas such as nitrogen or oxygen at low temperatures (when vibrational mode is not excited) D = 5, due to 3 translational degrees of freedom (linear motion in x,y and z directions) and 2 rotational degrees of freedom. Then the specific heat ratio of a diatomic gas can be calculated as

$$\gamma = \frac{5+2}{5} = \frac{7}{5} = 1.4$$

For a monatomic gas such as helium, argon, neon, etc. one can calculate the specific heat ratio as

$$\gamma = \frac{3+2}{3} = \frac{5}{3} = 1.67$$

since the molecule has only three translational degrees of freedom. Simplifying equations (5.10) and (5.11), it is easy to show that

$$h = u + pv = u + RT \rightarrow C_p T = C_v T + RT \Rightarrow C_p = C_v + R \tag{5.13}$$

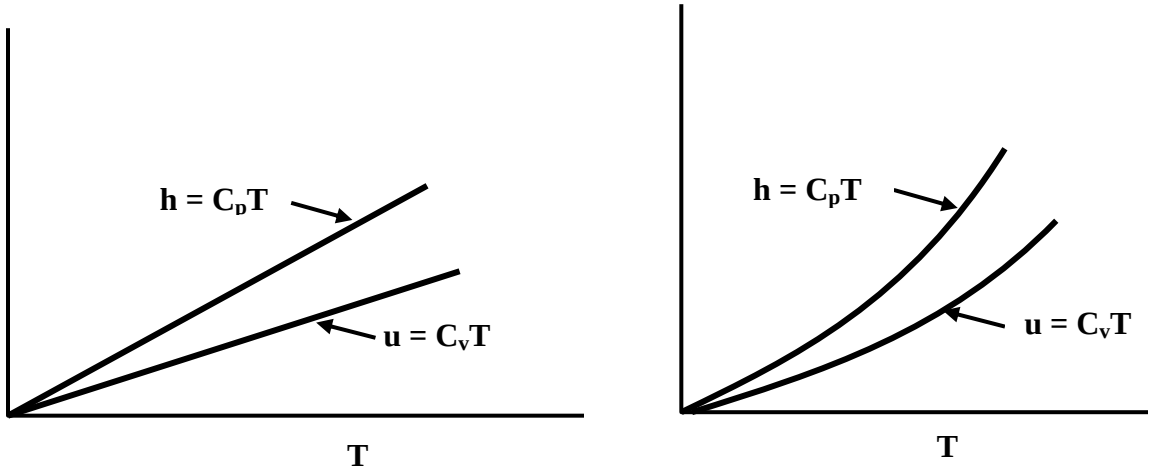
Now, further relations involving γ and R can be derived, for the specific heats.

$$\begin{aligned} &C_p = C_v + R \quad \rightarrow \gamma = \frac{C_p}{C_v} = 1 + \frac{R}{C_v} \quad \rightarrow \ \gamma - 1 = \frac{R}{C_v} \quad \rightarrow C_v = \frac{R}{\gamma - 1} \\ &\frac{C_p}{C_v} = \gamma \quad \rightarrow C_p = \frac{\gamma}{\gamma - 1} R \end{aligned} \tag{5.14}$$

Now let us apply the above relations for obtaining the specific heat values for nitrogen at low temperature. The molecular weight of nitrogen M = 28 kg/kmol. Then the specific gas constant R = 8314 (J/kmol K) /28 (kg/kmol) = 296.93 J/kg K. Since the specific

heat ratio γ = 1.4 (for degrees of freedom = 5), C_v = 296.93/(1.4-1.0) = 742.32 J/kg K and $C_p = \gamma.C_v$ =1.4 x 742.32 = 1039.25 J/kg K.

It was mentioned that the number of degrees of freedom may change with temperature because the vibrational mode gets excited at high temperatures. As temperature increases, the fraction of molecules with vibrational excitation will increase and this results in D (and so γ, as well as C_p and C_v) varying as functions of temperature. Ideal gases can be further classified into *Calorifically Perfect Gases* (or simply perfect gases) and *Calorifically Semi- Perfect Gases*. For perfect gases, the specific heats are constants and hence, the specific internal energy and specific enthalpy are linear functions of temperature. For semi-perfect gases, both u and h increase as non-linear functions of temperature- or, $C_p = C_p(T)$ and $C_v = C_v(T)$. Figure 5.1 shows these dependences schematically.



(a) Perfect gas (b) Semi- perfect gas

Fig. 5.1 Variations of specific internal energy and specific enthalpy for an ideal gas

5.3 Properties of Ideal Gas Mixtures

Let us consider a gaseous mixture which is made up of several ideal gas species. The composition of the mixture can be expressed in terms of the mass fractions or mole fractions of the individual species. Correspondingly, all the properties of the mixture

could presented in terms of the mass-based (specific properties) or mole-based (molar) properties of individual components.

Consider a mixture of N constituents. The number of moles of each species present in the mixture is denoted as n_1, n_2, n_3, …… n_N. The mass of individual species is denoted as m_1, m_2, m_3, ……. m_N. Now, the total number of moles 'n' and the total mass 'm' of the mixture are given by

$$n = n_1 + n_2 + n_3 + ... + n_N = \sum_{i=1,N} n_i; \qquad m = m_1 + m_2 + m_3 + ... + m_N = \sum_{i=1,N} m_i \quad (5.15)$$

The mole fraction x_i of any species *i* can be defined as

$$x_i = \frac{n_i}{\sum_{j=1,..N} n_j} = \frac{n_i}{n} \quad (5.16a)$$

Similarly, the mass fraction y_i of a species *i* can be defined as

$$y_i = \frac{m_i}{\sum_{j=1,..N} m_j} = \frac{m_i}{m} \quad (5.16b)$$

It is evident that the mass fractions and mole fractions should add up to unity, for the whole mixture. Thus,

$$\sum_{i-1,N} x_i = 1 \quad and \quad \sum_{i=1,N} y_i = 1 \quad (5.17)$$

It is easy to convert from the mole fraction to mass fraction, or vice versa. For example, consider one kmol of a mixture. Let the mole fractions for the individual components be x_1, x_2, …x_i,…x_N. The corresponding masses for the individual species can be found in terms of the respective molecular weights, as follows:

$$m_1 = x_1 \times M_1; \quad m_2 = x_2 \times M_2; ..... m_i = x_i \times M_i; ..... m_N = x_N \times M_N$$
$$m = \sum_{i=1,N} m_i = \sum_{i=1,N} x_i M_i \quad (5.18)$$

The mass fraction y_i is now found in terms of the mole fractions, as follows:

$$y_i = \frac{m_i}{m} = \frac{x_i M_i}{\sum_{i=1,N} x_i M_i} \tag{5.19}$$

Since one kmol was considered in the discussion above, the corresponding mass of the mixture is nothing but the molecular weight of the mixture. Thus, the mixture molecular weight M is given as

$$\sum_{i=1,N} x_i M_i = M \tag{5.20}$$

Now, in order to find the inverse relationship between the mole fractions and mass fractions, let us consider one kg of gas mixture. The masses of the individual species in kilograms are equal to y_1, y_2, …y_i,…y_N. The corresponding number of kmols are given by:

$$\begin{aligned} &n_1 = \frac{y_1}{M_1}; \quad n_2 = \frac{y_2}{M_2}; \ldots\ldots\ldots n_i = \frac{y_i}{M_i}; \ldots\ldots n_N = \frac{y_N}{M_N} \\ &n = \sum_{i=1,N} n_i = \sum_{i=1,N} \frac{y_i}{M_i} \end{aligned} \tag{5.21}$$

Thus, the mole fraction x_i can be found in terms of the mass fractions as:

$$x_i = \frac{n_i}{n} = \left(\frac{y_i}{M_i} \right) / \left(\sum_{j=1,N} \frac{y_j}{M_j} \right) \tag{5.22}$$

In a mixture of ideal gases, each component can be assumed to still behave as an ideal gas. In order to describe the state (pressure, volume, temperature, etc.) of each component, however, there are two equivalent models- the Dalton's law of partial pressures and the Amagat's law of partial volumes. As shown schematically in Fig. 5.2a, the Dalton's law proposes that each component *i* can be assumed to exert a partial pressure p_i at the mixture temperature and volume (T,V), such that the sum of all partial pressures is equal to the total mixture pressure p. The Amagat's law visualizes

that at mixture pressure and temperature (p,T), each component *i* occupies a partial volume V_i such that the sum of all partial volumes is equal to the total mixture volume V. This visualization is schematically depicted in Fig. 5.2b. In other words, an ideal gas component *i* can be assumed to be at the condition of p_i, V, T in the mixture as per the Dalton's law; it can be assumed to be at the condition of p, V_i, T as per the Amagat's law.

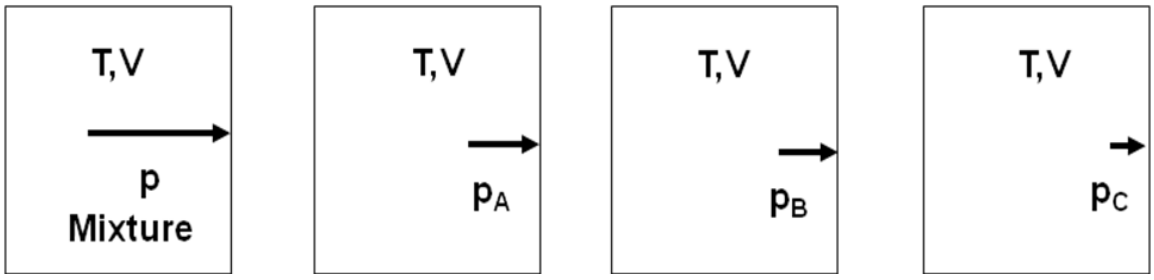


Fig. 5.2a State of each component as per Dalton's law

Dalton's law: Sum of all partial pressures = total mixture pressure. Mathematically,

$$\sum_{i=1,N} p_i = p \qquad (5.23a)$$

Amagat's Law: Sum of all partial volumes = total volume of mixture. Therefore,

$$\sum_{i=1,N} V_i = V \qquad (5.23b)$$

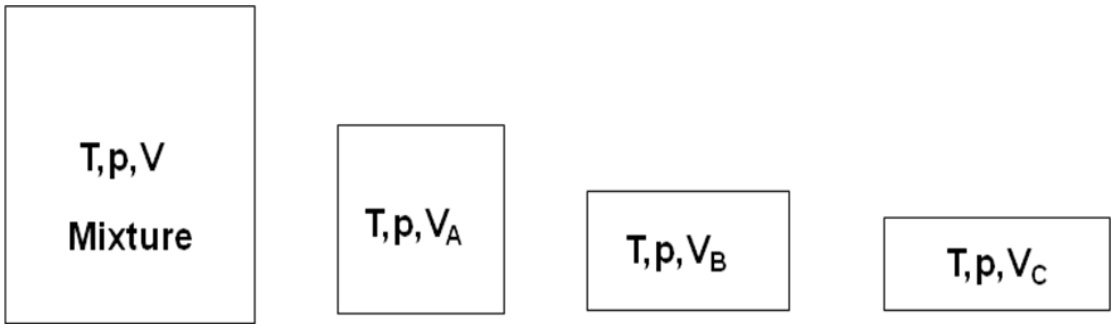


Fig. 5.2b State of each component as per Amagat's law

The above two models are equivalent and they are based on the fact that each component in the mixture still behaves as an ideal gas, even when other gases are present.

Now using the Dalton's law, the state of each gas can be described through the following relationships:

$$p_1 V = n_1 \overline{R} T$$
$$p_2 V = n_2 \overline{R} T$$
$$\ldots\ldots\ldots\ldots\ldots$$
$$p_i V = n_i \overline{R} T \quad for\ any\ i$$
$$\ldots\ldots\ldots\ldots\ldots$$

For the mixture $pV = n\overline{R}T$

It is evident from the above relations that $\frac{p_i}{p} = \frac{n_i}{n} = x_i$ (5.24a)

Similarly, from Amagat's law, we get:

$$pV_1 = n_1 \overline{R} T$$
$$pV_2 = n_2 \overline{R} T$$
$$\ldots\ldots\ldots\ldots\ldots$$
$$pV_i = n_i \overline{R} T \quad for\ any\ i$$
$$\ldots\ldots\ldots\ldots\ldots$$

For the mixture $pV = n\overline{R}T$

Dividing the equation for species *i* by that of the mixture gives:

$$\frac{V_i}{V} = \frac{n_i}{n} = x_i \qquad (5.24b)$$

Thus, the values of partial pressures or partial volumes needed to describe the state of a component in an ideal gas mixture, can be determined in terms of the corresponding mole fractions. Now, all the properties of the gaseous mixture can be evaluated from the component level values using mass-basis approach or a molar-basis approach. We can employ specific properties of the mixture or molar properties of the mixture for this purpose.

Let us consider any extensive property such as volume, internal energy or enthalpy. For instance, the internal energy U for the mixture can be expressed in mass-basis terms as follows:

$$U = m \times u = m_1 u_1 + m_2 u_2 + ...m_i u_i + ...m_N u_N = \sum_{i=1,N} m_i u_i \qquad (5.25)$$

Dividing by the mass of the mixture, we get:

$$u = \frac{U}{m} = \sum_{i=1,N} \left(\frac{m_i}{m} \right) u_i = \sum_{i=1,N} y_i u_i \qquad (5.26a)$$

Similarly it can be shown that the specific enthalpy h and the specific volume v of the mixture are given as:

$$h = \frac{H}{m} = \sum_{i=1,N} \left(\frac{m_i}{m} \right) h_i = \sum_{i=1,N} y_i h_i \qquad (5.26b)$$

$$v = \frac{V}{m} = \sum_{i=1,N} \left(\frac{m_i}{m} \right) v_i = \sum_{i=1,N} y_i v_i \qquad (5.26c)$$

In the above expressions, u_i, h_i and v_i are respectively the specific internal energy, specific enthalpy and the specific volume of each component *i* in the mixture.

Using the molar basis, we can similarly derive the following relations:

$$U = n \times \overline{U} = n_1 \overline{U}_1 + n_2 \overline{U}_2 + ...n_i \overline{U}_i + ...n_N \overline{U}_N = \sum_{i=1,N} n_i \overline{U}_i \qquad (5.27)$$

Dividing by the number of mole of the mixture gives

$$\overline{U} = \frac{U}{n} = \sum_{i=1,N} \left(\frac{n_i}{n} \right) \overline{U}_i = \sum_{i=1,N} x_i \overline{U}_i \qquad (5.28a)$$

Similarly,

$$\overline{H} = \frac{H}{n} = \sum_{i=1,N} \left(\frac{n_i}{n} \right) \overline{H}_i = \sum_{i=1,N} x_i \overline{H}_i \qquad (5.28b)$$

and

$$\overline{V} = \frac{V}{n} = \sum_{i=1,N}\left(\frac{n_i}{n}\right)\overline{V_i} = \sum_{i=1,N} x_i \overline{V_i} \qquad (5.28c)$$

It is evident from Eqs. (5.26a-c) that the specific properties of the ideal gas mixture can be expressed as the product of species mass fraction and the specific property of each component, summed over all the species. Similarly, a molar property of the mixture can be expressed as the product of species mole fraction and the molar property of each component, summed over all the species.

Example 1: Consider an ideal gas mixture of 1 kmol of hydrogen and 1 kmol of nitrogen. Find the mass fractions and mole fractions of hydrogen and nitrogen. Also find the molecular weight, specific gas constant R and the specific heats C_p and C_v of the mixture. If this mixture is heated from an initial temperature of 300 K to the final temperature of 400 K at constant pressure, find: Δu, Δh, Q and W for the process.

Molecular weight of hydrogen = 2 kg/ kmol.

Molecular weight of nitrogen = 28 kg/ kmol.

Total mass (m) of the mixture = 1 kmol x 2 kg/kmol + 1 kmol x 28 kg/kmol = 30 kg.

Molecular weight of the mixture, M = 30 kg/ 2 kmols = 15 kg/kmol.

Mass fraction of hydrogen, y_{H2} = 2 kg/ 30 kg = 1/15.

Mass fraction of nitrogen, y_{N2} = 28 kg/ 30 kg = 14/15.

Mole fraction of hydrogen, x_{H2} = 1 kmol/ 2 kmols = 0.5.

Mole fraction of nitrogen, x_{N2} = 1 kmol/ 2 kmols = 0.5.

Specific gas constant R = 8314/15 = 554.27 J/kg K.

Since both hydrogen and nitrogen are diatomic gases, their specific heat ratio = 1.4 and so the specific heat ratio for the mixture γ = 1.4.

Specific heat at constant volume for the mixture $C_v = R/(\gamma-1) = 554.27/0.4 = 1385.67$ J/kg K. The value of $C_p = 1.4 \times 1385.67 = 1939.93$ J/kg K.

The change in specific internal energy $\Delta u = C_v \times \Delta T = 1385.67 \times (400 - 300) = 138567$ J/kg = 138.567 kJ/kg.

The change in specific enthalpy $\Delta h = C_p \times \Delta T = 1939.93 \times (400 - 300) = 193.993$ kJ/kg.

For the constant pressure process, $Q = \Delta H = 30$ kg x 193.933 = 5818 kJ.

Change in internal energy $\Delta U = 30$ kg x 138.567 = 4157 kJ.

Work, $W = Q - \Delta U = 5818 - 4157 = 1661$ kJ.

5.4 Properties of Steam/ Water substance

Substances are known to exist in three distinct phases- namely, the solid, liquid and vapor phases. The pressure, temperature and volume conditions and the properties such as internal energy, enthalpy and specific heats for the different phases could be widely different. For example, although water and steam are chemically the same (H_2O), under the same temperature and pressure conditions, liquid water could be about 1000 times denser than steam! Since water is an abundantly available substance and it is used as a working substance in most of the power plants, we shall study the variation of its properties corresponding to different phases. In particular, we consider in detail the properties corresponding to the vapor phase (steam) and the liquid phase (water).

Let us consider two experiments with water, namely- the constant pressure heating and the constant temperature expansion, in order to understand the changes in its properties.

(i) Constant pressure heating of water:

Let us consider a vertical piston- cylinder arrangement, where the pressure inside the cylinder can be maintained as desired with the help of an appropriate mass placed on the piston (see Fig. 5.3). Now let the contents of the cylinder be heated slowly. Let us

assume that initially only water is present. As heat is added to the water, its temperature starts increasing with time. Correspondingly, the volume also increases. When the liquid temperature becomes equal to the boiling point temperature, bubbles appear because of steam formation. During this boiling phase, the water/steam temperature remains constant, along with pressure. The variation of temperature with volume during this constant pressure heating process (1-f-g-2) is shown in Fig. 5.3.

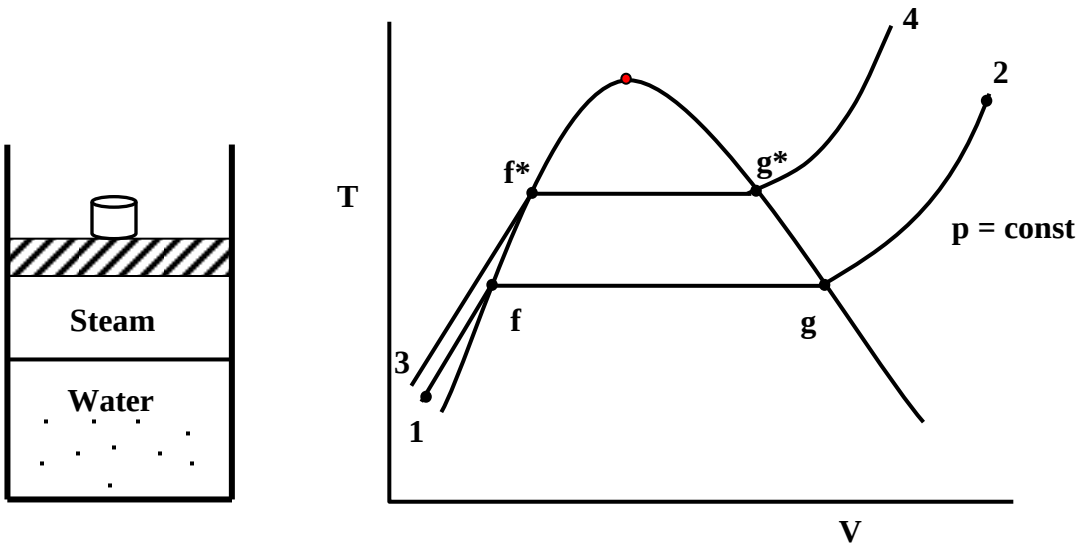


Fig. 5.3 Constant pressure heating of water

Here, 'f' represents the **saturated liquid** state and 'g' denotes the **saturated vapor** state at the same pressure. During the portion f-g of the process, the entire liquid is converted into steam, through boiling. Now, let us consider the situation when the pressure is increased by adding some more mass above the piston. The corresponding constant pressure heating process is shown as another curve 3-f*-g*-4, similar to the process 1-f-g-2. The line connecting all saturated liquid states f,f*, etc at different pressures, is called the saturated liquid line. Similarly, the line connecting all saturated vapor states g,g*, etc is known as the saturated vapor line. The region in between the saturated liquid and saturated vapor lines, corresponds to the states of saturated steam and water mixture in different proportions. For instance, state 'f' denotes 100% of saturated liquid while 'g' denotes 100% of saturated vapor. If a state 'i' lies between f and g, it corresponds to a mixture of saturated steam and saturated water (Fig. 5.4). For the mixture states, we employ a property x called as quality or dryness fraction. We define this property as:

Quality (or dryness fraction) x = mass of vapor/ mass of total mixture.

Thus, if x_i is the quality at the state 'i', then in one kg of mixture at that state, the amount of vapor present will be x_i kg and the amount of saturated water will be $(1-x_i)$ kg. Every extensive property at state 'i' can be expressed in terms of the quality. For example, properties of the mixture at state 'i' such as the specific volume, specific internal energy and specific enthalpy can be expressed as:

$$\begin{aligned} v_i &= x_i \times v_g + (1-x_i) \times v_f \\ u_i &= x_i \times u_g + (1-x_i) \times u_f \\ h_i &= x_i \times h_g + (1-x_i) \times h_f \end{aligned} \qquad (5.29)$$

The above expressions illustrate that the property of the mixture is calculated as: vapor fraction x saturated vapor property + liquid fraction x saturated liquid property. Thus, the dryness fraction or quality is a very important property for evaluating all the other extensive property values of the mixture. The saturated state properties (v_f, v_g, u_f, u_g etc) can be evaluated from saturated steam tables. There are two tables which provide these properties- the saturated steam temperature table and the saturated steam pressure table. For example, at a temperature of 100°C, from the temperature table, we find that v_f = 1.044 x 10^{-3} = 0.001044 m^3/kg and v_g = 1.673 m^3/kg. Similarly, specific internal energy values are: u_f = 418.9 kJ/kg u_g = 2507 kJ/kg; specific enthalpy values are: h_f = 419 kJ/kg and h_g = 2676 kJ/kg. Suppose a steam water mixture has a quality of 30% at 100°C in state 'i' , then the mixture properties can be calculated using Eqs. (5.29) as:

$$\begin{aligned} v_i &= 0.3 \times 1.673 + (1-0.3) \times 0.001044 = 0.5026\ m^3/kg\ . \\ u_i &= 0.3 \times 2507 + (1-0.3) \times 418.9 = 1045.33\ kJ/kg \\ h_i &= 0.3 \times 2676 + (1-0.3) \times 419 = 1096.1\ kJ/kg. \end{aligned}$$

In a similar manner, if a mixture state 'i' is specified in terms of the pressure and quality as p_i = 3 bar and x_i = 0.4, then the properties can be evaluated using the saturated pressure table as follows:

$$v_i = 0.4 \times 0.606 + (1 - 0.4) \times 0.001073 = 0.243\, m^3/kg\,.$$
$$u_i = 0.4 \times 2544 + (1 - 0.4) \times 561.1 = 1354.3\, kJ/kg$$
$$h_i = 0.4 \times 2725 + (1 - 0.4) \times 561.5 = 1426.9\, kJ/kg.$$

Depending on the properties specified for the saturated mixture state, one can employ either the saturated temperature table or the saturated pressure table, as the case may be. Obviously, the above procedure can be used only for mixture states lying in the dome region (in between the saturated liquid and the saturated vapor states). The top most point of the dome is indicated in red colour in Fig. 5.3. This state is known as the 'critical point' or the 'critical state'. For the saturated steam tables, it can be seen that the critical state corresponds to a pressure of 220.9 bar and a temperature of 374.14 ^{o}C. The critical state is defined as the state at which distinction between the liquid and vapor states is lost. For example, in a glass container if a mixture of saturated water and saturated steam is kept at a pressure of 1 atm (100°C), we will be able to distinguish the water and steam regions through the interface which separates the two phases. This is because of the fact that the densities of liquid water and steam are vastly different at this pressure, and refraction of light through these two media will clearly indicate the phase boundary. On the other hand, at critical state, the densities of both the liquid and the vapor phases are the same and we will not be able to distinguish the liquid from the vapor.

With reference to the state diagram drawn with T and V (or specific volume v) as the coordinates (Fig. 5.4), the following features can be observed. The saturated liquid line and the saturated vapor line meet at the critical point. The states to the left of the saturated liquid line correspond to the 'compressed liquid' or 'sub-cooled liquid' states. The states lying in between the saturated liquid and the saturated vapor lines (under the dome') are the saturated mixture states. The states lying to the right of the saturated vapor line correspond to the superheated steam states. For the saturated mixture states, the pressure and temperature are not independent properties. For example, if the pressure value is 1 bar, then automatically the corresponding boiling point temperature (known as the saturation temperature T_{sat}) is equal to 99.62^{o}C; if the saturated mixture temperature is given as 120°C, the corresponding saturation pressure

p_{sat} is equal to 1.985 bar (as seen from the saturated steam pressure table or temperature table). If a state, lies in the 'superheated steam' region, then one needs to refer to the superheated steam tables. For example, At a pressure of 8 bar and temperature of 300°C, we find that v = 0.324 m^3/kg, u = 2797 kJ/kg and h = 3057 kJ/kg for the superheated steam. As seen from Fig. 5.3, at any pressure p, all states with $T > T_{sat}$ will correspond to superheated steam; also, states with $T < T_{sat}$ will correspond to sub-cooled (or compressed) liquid states.

In Fig. 5.3, we illustrated the consequences of adding heat at constant pressure to liquid water. Suppose a similar experiment dealing with the expansion at constant temperature is conducted (this could be performed by immersing the piston- cylinder device into a constant temperature bath and moving the piston upwards as shown in Fig. 5.4), the variation of pressure with volume at constant temperature can be plotted as shown in Fig. 5.5. Let us consider expansion at a constant temperature of 50°C, indicated by the curve 1-f-g-2. In the compressed liquid regime 1-f, a slight increase in volume gives rise to a drastic reduction in the pressure. Eventually, when the saturated liquid state 'f' is reached, water starts boiling at 50°C if the mixture is expanded further. When all the water has just evaporated at this temperature, the saturated vapor state g is attained. Further expansion, results in superheated steam at 50°C. This example shows the plotting of isotherms (constant temperature curves) in the p-v diagram.

In reality, the T-v diagram or p-v diagram is a projection (plan view or the elevation view) of the general p-v-T surface of a substance. For substances that contract on solidification or expand on solidification, the p-v-T surfaces including the solid, liquid and vapor regions are shown in Fig. 5.6. As seen in the figure, transition from solid to liquid occurs through a mixture region consisting of both the solid and liquid, by the process of melting. Transition from the liquid to vapor occurs through a mixture of liquid and vapor through the process of boiling. At low pressures, transition from the solid to vapor can occur directly through the solid-vapor mixture region by the process of sublimation. A particular state having the pressure p_{TP} and temperature T_{TP} , known as the 'Triple point' exists at which all the three states of matter (solid, liquid and vapor) can be in equilibrium with each other. In Fig. 5.6 this state is shown as the triple line, corresponding to different amounts of solid, liquid and vapor.

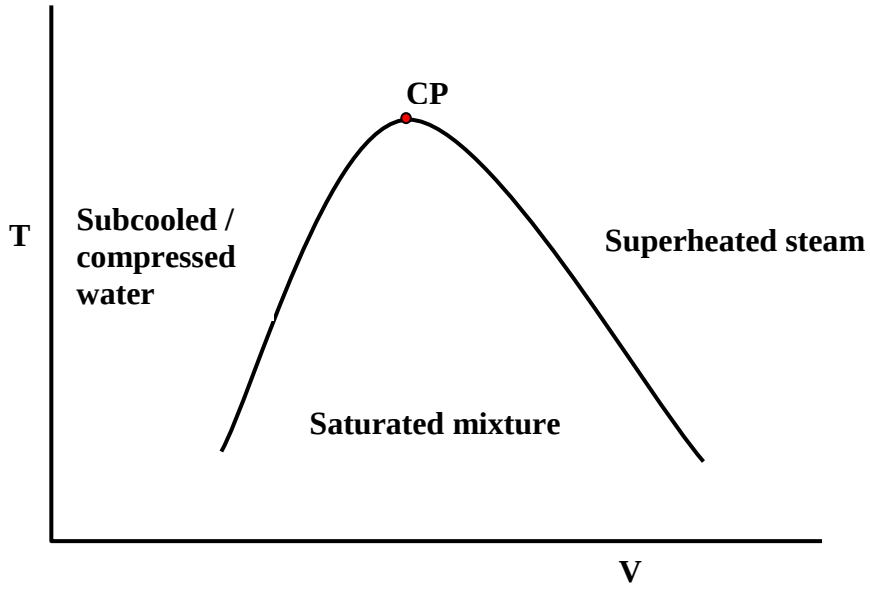


Fig. 5.4 Liquid, vapour and mixture states

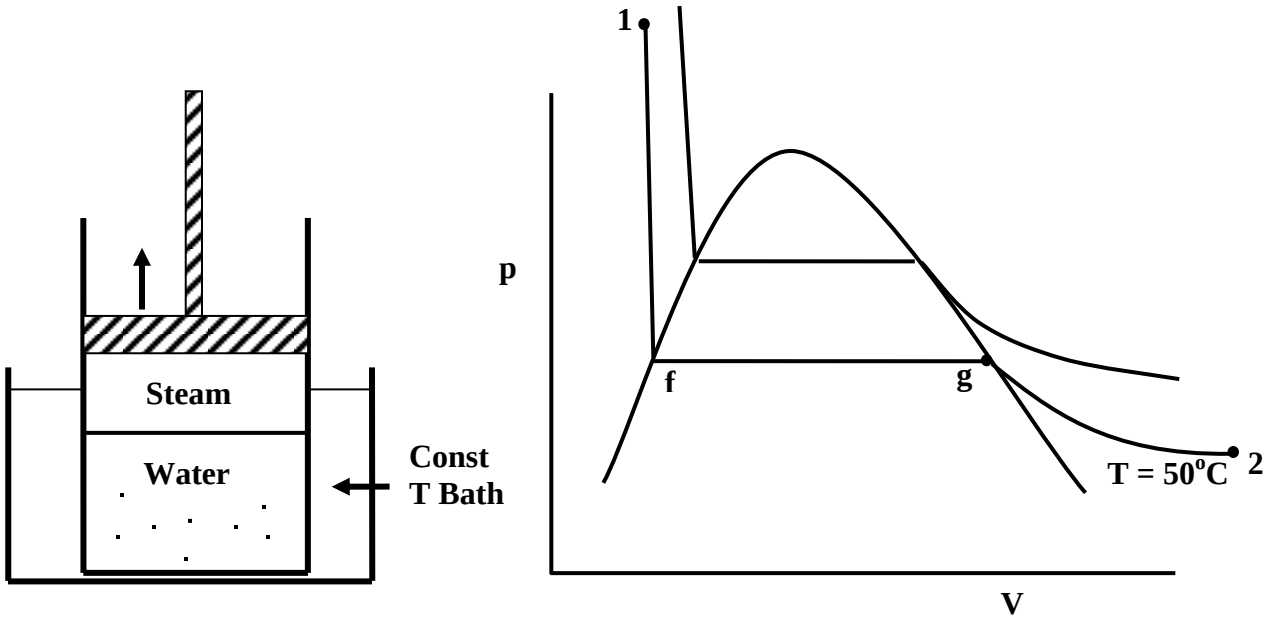


Fig. 5.5 Constant temperature expansion

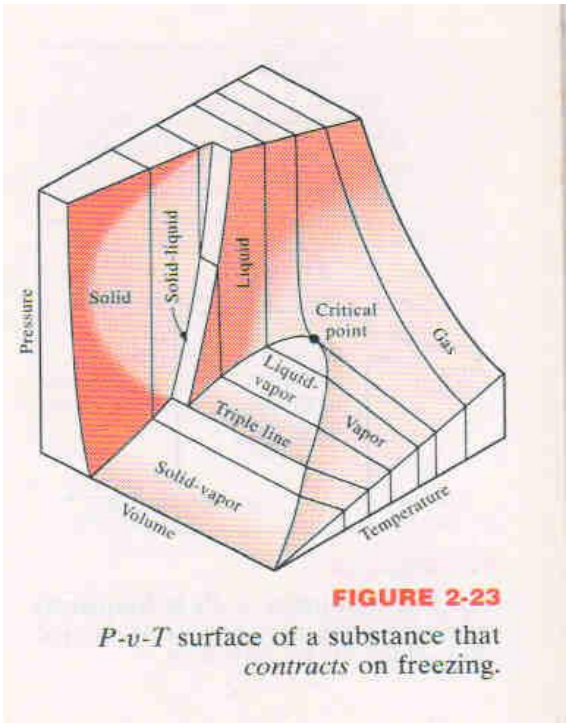


FIGURE 2-23

P-v-T surface of a substance that *contracts* on freezing.

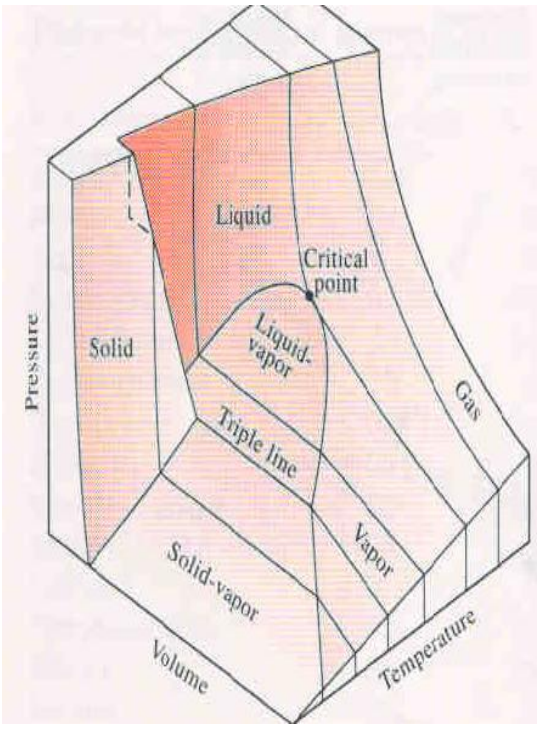


Fig. 5.6 p-v-T surface for substances which contract or expand during freezing

Chapter 6 I Law for Control Volume

6.1 Introduction

In Chapter 1, **'Control Volume'** was defined as a fixed volume on which attention is focused. Apart from energy interactions in the form of heat or work with the surroundings, it can also admit in-flow (m_i) or out-flow of mass (m_e) through inlet or exit, respectively. Practical devices involving the flow of a fluid, such as a turbine, pump, compressor, nozzle, valve, boiler, condenser, etc. can be modeled as CV. The form of I Law for a Control Volume is different from that of a system, on account of the fact that there is mass exchange. Moreover, in addition to the internal energy, forms of energy storage such as the kinetic energy and potential energy could be important in many cases.

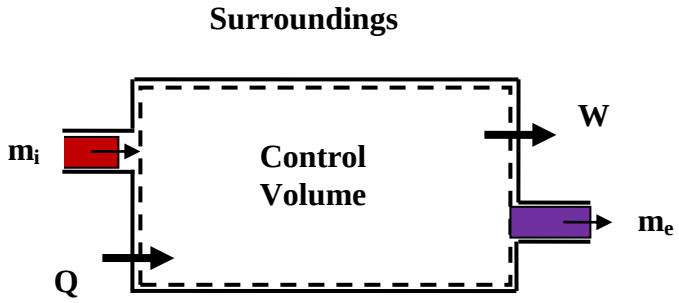


Fig. 6.1 Control Volume Analysis

Let us now turn our attention to the modes of work interaction. For a control volume, the work interaction W cannot include displacement work (i.e. work of expansion or compression). This is because of the fact that the volume of CV remains constant. Other forms such as electrical work, magnetic work, shaft work etc. could be included in W. Some authors call this W as 'external work', implying that there is also some internal work within the fluid. For instance, the incoming mass m_i pushes the fluid ahead of it, like a piston. Similarly, the outgoing mass m_e also pushes the fluid ahead of it. This form of work is known as the 'flow work' and it is given by the expression (m.p.v), where p and v are the local pressure and the local fluid specific volume. More explanation on flow work is provided subsequently.

6.2 Derivation of I Law for a CV

Let us now derive the energy balance (I law) principle for a control volume. For the sake of simplicity, let us consider a single inlet and single outlet situation, although the energy balance can be generalized later on, to the case of multiple inlets and outlets. For obtaining the I law for CV, we shall use the familiar I law for a system over a small time interval Δt. In other words, we shall consider two time instants 't' and 't+Δt', when a system of given mass occupies the control volume as shown in Fig. 6.2.

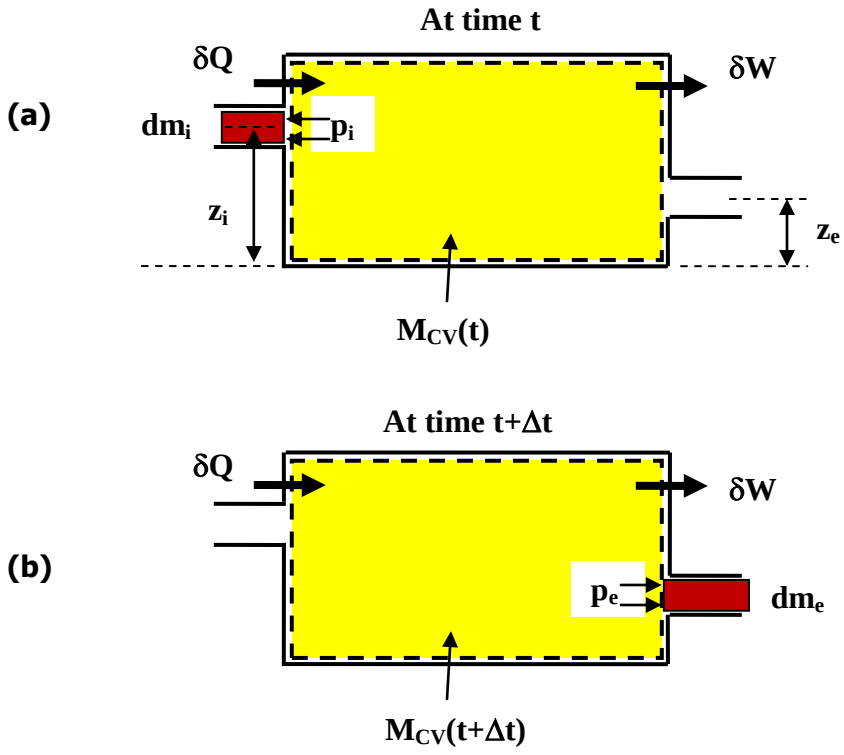


Fig. 6.2 Energy balance for fixed mass over time interval Δt

The same mass (Fig. 6.2 a) which consists of dm_i + $M_{CV}(t)$ at time 't', occupies the configuration dm_e + $M_{CV}(t+\Delta t)$ at time 't+Δt' (Fig. 6.2 b). Here, dm_i and dm_e are the masses that enter into or exit from the control volume over the time interval Δt. Also, $M_{CV}(t)$ and $M_{CV}(t+\Delta t)$ are the masses that occupy the control volume at time 't' and time 't+Δt' respectively.

Now, the total energy of the system at time 't' is equal to :

$$E_{CV}(t) + dm_i \left(u_i + \frac{V_i^2}{2} + g\, z_i\right)$$

where $E_{CV}(t)$ is the total energy (internal + kinetic + potential energy) stored in the control volume at time 't'. Also, the second term represents the total energy (internal + kinetic + potential energy) brought by the incoming mass dm_i. Here, V_i is the inlet flow velocity and z_i is the elevation of the inlet. Similarly, the total energy of the system at time 't+Δt' can be written as

$$E_{CV}(t + \Delta t) + dm_e \left(u_e + \frac{V_e^2}{2} + g\, z_e\right).$$

The first and second terms of the above expression represent the energy stored in the CV at time 't+Δt' and the total energy carried away by the out-going mass dm_e.

Now, applying I law for the system (fixed mass under consideration), we can write:

Heat added – Work done = Change in total energy.

The work interaction consists of the work δW done across the CV boundary and also the flow work at the inlet and the flow work at the exit. The flow work at the inlet is negative, because fluid which is flowing towards the CV is doing work on the system under consideration. This work can be evaluated as:

Flow work at the inlet = - p_i . dV_i .

Here dV_i is the volume of fluid flowing against the resisting pressure of p_i at the inlet (Fig. 6.2 a), within the time interval of 't' and 't+Δt' . The volume dV_i can be calculated as $dm_i.v_i$, where v_i is the specific volume of fluid in the inlet region. Thus, the flow work at inlet can be expressed as – dm_i (p_iv_i). Similarly, the flow work at the outlet is given as + dm_e (p_ev_e). Now, application of I law for the system under consideration gives:

$\delta Q - \{\delta W + dm_e\ (p_e v_e) - dm_i\ (p_i v_i)\} = \{E_{CV}(t+\Delta t) + dm_e\,(u_e + \frac{V_e^2}{2} + g\,z_e)\}$ -

$\{E_{CV}(t) + dm_i\,(u_i + \frac{V_i^2}{2} + g\,z_i)\}$

After rearranging the terms, we get:

$$\delta Q - \delta W = E_{CV}(t+\Delta t) - E_{CV}(t) + dm_e\{u_e + p_e v_e + \frac{V_e^2}{2} + g\,z_e\}$$
$$- dm_i\{u_i + p_i v_i + \frac{V_i^2}{2} + g\,z_i\}$$

Since specific enthalpy h = u + pv, this expression can be simplified as

$$\delta Q - \delta W = E_{CV}(t+\Delta t) - E_{CV}(t) + dm_e\{h_e + \frac{V_e^2}{2} + g\,z_e\} - dm_i\{h_i + \frac{V_i^2}{2} + g\,z_i\}$$
$$= dE_{CV} + dm_e\{h_e + \frac{V_e^2}{2} + g\,z_e\} - dm_i\{h_i + \frac{V_i^2}{2} + g\,z_i\}$$

Now, dividing the above energy balance expression by the time interval Δt and taking the limit as $\Delta t \rightarrow 0$, results in the equation

$$\dot{Q} - \dot{W} = \frac{dE_{CV}}{dt} + \dot{m}_e\{h_e + \frac{V_e^2}{2} + g\,z_e\} - \dot{m}_i\{h_i + \frac{V_i^2}{2} + g\,z_i\}. \qquad (6.1)$$

The terms on the left hand side of Eq. (6.1) refer to the rate of heat addition and the rate of work done across the control volume boundary. The first term on the right hand side refers to the rate of total energy storage in the CV. The second term on RHS includes the rate of total energy carried by the out flow and the rate of flow work done at the exit. Remember that enthalpy h includes the internal energy 'u' and the flow work 'pv' per kg of flow. The last term on RHS represents the rate of total energy brought by

the inflow and the rate of flow work done at the inlet. In the case of a problem with multiple inlets and exits, the equation (6.1) can be modified to a more general form

$$\dot{Q} - \dot{W} = \frac{dE_{CV}}{dt} + \sum_e \dot{m}_e \{h_e + \frac{V_e^2}{2} + g\, z_e\} - \sum_i \dot{m}_i \{h_i + \frac{V_i^2}{2} + g\, z_i\}. \quad (6.2)$$

Here the summations are taken over all the exits 'e' and the inlets 'i'. The applications of I law for a CV can be brought under two major classifications, namely, those of steady flow devices and unsteady flow devices. In the case of a thermal power plant which carries out energy conversion from heat to work, many of the sub-components can be modeled as Control Volumes, as discussed below. The steam generator or boiler, takes in heat input and converts water into steam at high pressure and temperature. A steam turbine is a device in which the high pressure steam impinges on the blades and produces rotary motion (i.e. generates shaft work). In the condenser, steam (after transferring part of its energy as work in the turbine) is condensed back to water. A pump raises the pressure of the condensed water and sends it back to the steam generator (Fig. 6.2 a). Since a power plant operates in a continuous mode, except for some occasional shut-downs, it is meaningful to analyze the associated components as steady flow devices.

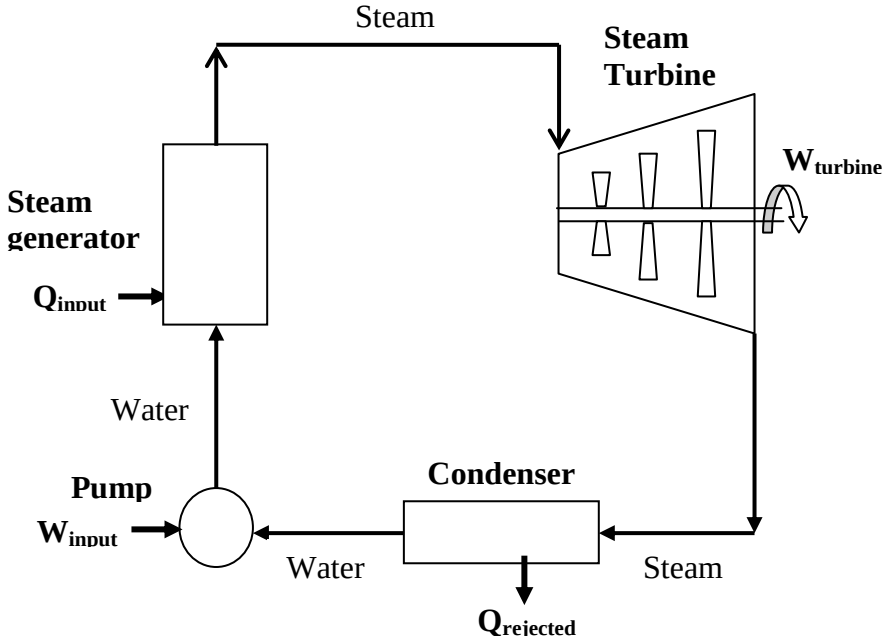


Fig. 6.2 a Schematic of Thermal Power Plant

In a typical aero-engine, atmospheric air which enters at a high relative velocity (because of the high speed of the aircraft) is slowed down in a diffuser, which converts the kinetic energy into pressure. The air is further compressed in a compressor and sent to the combustor where it burns with a fuel and produces heat. The hot products of combustion expand in a turbine producing mechanical (shaft) work. Finally, the hot combustion products are further expanded in a nozzle, generating a high speed jet exhaust. The high momentum jet generates forward thrust according to Newton's III law (Fig. 6.2 b). During the continuous operation of the engine, steady flow analysis could be carried out for each component, using the I law for a CV.

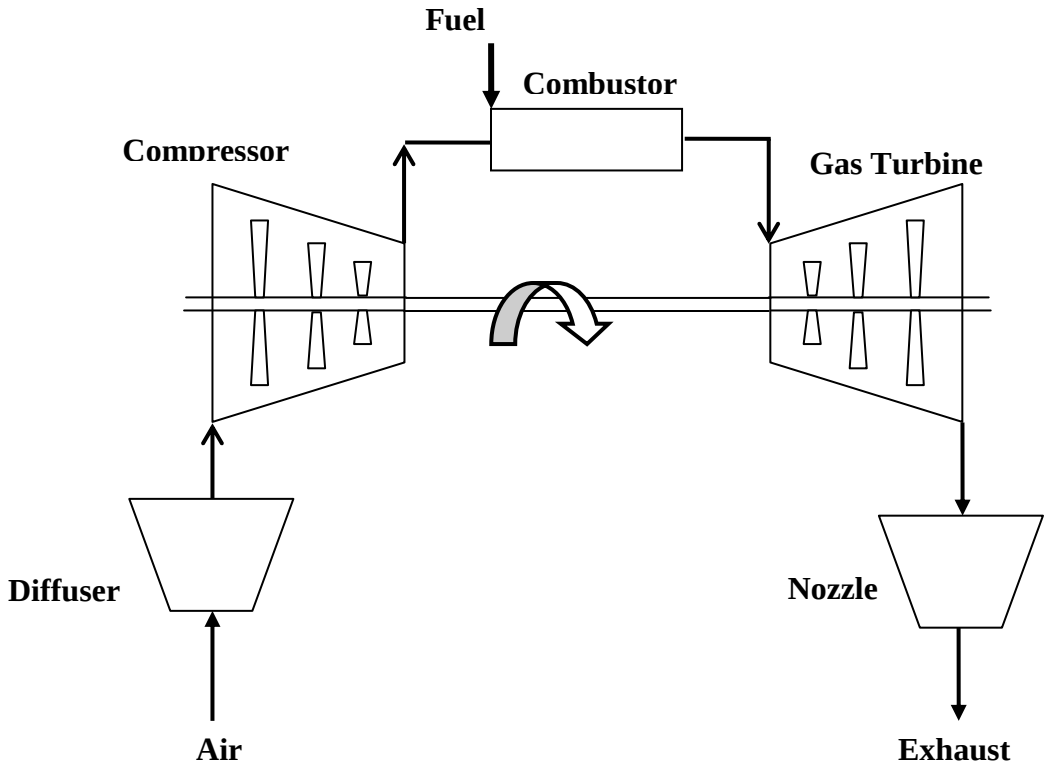


Fig. 6.2 b Schematic of Aero-engine system

6.3 Steady Flow Devices

Let us now consider the steady flow operation of various devices. Under the steady flow assumption, accumulation of mass or energy within the CV is taken to be zero. In other words, at every instant, the rate of inflow of mass to the CV is equal to the rate of out-flow from the CV. Also, the rate at which energy enters the CV (as heat, work or energy

of the flowing mass) is equal to the rate at which it leaves the control volume. Thus for steady state, we can set

$$\frac{dE_{CV}}{dt} = 0$$

With this simplification, equation (6.1) for a single inlet- single outlet device can be rewritten as:

$$\dot{Q} - \dot{W} = \dot{m}_e\{h_e + \frac{V_e^2}{2} + g\, z_e\} - \dot{m}_i\{h_i + \frac{V_i^2}{2} + g\, z_i\}$$

$$= \dot{m}\{h_e - h_i + \frac{V_e^2 - V_i^2}{2} + g(z_e - z_i)\} \qquad (6.3)$$

$$where \;\; \dot{m} = \dot{m}_e = \dot{m}_i = mass \;\; flow \; rate \; through \; the \; device.$$

Now let us look at further simplifications for specific devices.

(i) Adiabatic devices:

We came across devices such as steam turbine, gas turbine, compressor, pump, nozzle, diffuser, etc. in our discussions about the steam power plant and the aero-engine. In ideal situations, these devices are supposed to be adiabatic. For example, an ideal steam turbine is supposed to convert the energy contained in steam into useful mechanical work, without any heat losses to the surroundings. Therefore, we can set rate of heat transfer = 0 for the turbine. Furthermore, for any device dealing with gas flow, the potential energy difference can be neglected. (The potential energy term will be important only for devices handling liquid flows, especially when the height differences are significant.) Thus, we can neglect the term of $g.(z_e - z_i)$. In many cases, even the kinetic energy term may be negligible. Then I law for a steam turbine simplifies as

$$-\dot{W} = \dot{m}(h_e - h_i) \Rightarrow \dot{W} = \dot{m}(h_i - h_e) \qquad (6.4a)$$

The above expression implies that the power developed by the turbine is equal to the product of steam mass flow rate and the decrease of steam enthalpy from the inlet to

the outlet. When the kinetic energy change is important, the I law for the turbine can be written as

$$-\dot{W} = \dot{m}\{(h_e - h_i) + \frac{V_e^2 - V_i^2}{2}\} \qquad (6.4b)$$

Equations (6.4a) or (6.4b) are valid also for a gas turbine, which operates with the combustion product gas as working fluid. One can assume the product gas to be an ideal gas and calculate the properties using ideal gas relations. The steam turbine and gas turbine are positive work producing devices. The expressions (6.4a) and (6.4b) can be used for the analyzing the compressor also. In the example shown in Fig. 6.2b, the compressor compresses the air (air density, pressure and temperature will increase because of compression) by taking work input from the turbine. Just as a compressor increases the pressure of a gas, a pump will increase the pressure of a liquid with the help of work input. The increase in pressure will result in the increase of enthalpy from the inlet to the outlet of the pump. Thus, Eq. (6.4a) is valid for a pump also. Normally, the change in kinetic energy can be neglected for a pump, because of low velocities. Also, the change in potential energy can be neglected because the height differences between the inlet and outlet are usually small.

From the above discussion, it is clear that W> 0 for a turbine, while W < 0 for a pump or a compressor. In adiabatic devices such as nozzle or diffuser, heat as well as work interactions are zero. Consider an adiabatic nozzle with steady flow, as shown in Fig. 6.5a.

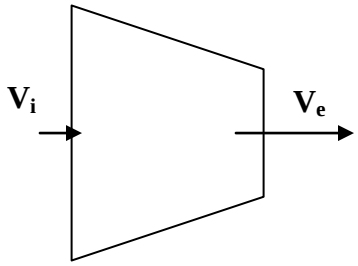


Fig. 6.5a Adiabatic nozzle

Since both heat transfer and work done are zero, the I law simplifies to the form:

$$h_i + \frac{V_i^2}{2} = h_e + \frac{V_e^2}{2} \tag{6.5}$$

Often, the inlet kinetic energy may be neglected and this leads to the simplification

$$V_e = \sqrt{2(h_i - h_e)} \tag{6.6}$$

for the nozzle exit velocity. For steady flow, we can also apply the mass balance principle as follows:

$$\dot{m}_i = \rho_i V_i A_i = \dot{m}_e = \rho_e V_e A_e = \dot{m} \tag{6.7}$$

The above expression implies that the mass flow rates at the inlet and the outlet are both equal to the overall flow rate of $\dot{m}$ through the nozzle. Also, the mass flow rate is given as the product of the density, velocity and flow area at the inlet or outlet. It is evident from the above discussion that the purpose of a nozzle is to accelerate a flow (increase kinetic energy) at the cost of its enthalpy. A diffuser does exactly the opposite function as that of a nozzle. Flow will enter a diffuser at higher kinetic energy and leave with a lower kinetic energy, with the difference being converted into flow enthalpy. In effect, the pressure and temperature of the flow will increase because of the conversion from kinetic energy to enthalpy.

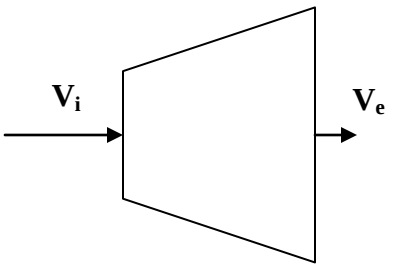


Fig. 6.5b Adiabatic diffuser

Finally, we consider the device of throttle valve which is used for controlling the flow rates in pipe lines and other fluid equipment. The heat and work interactions for a throttle valve may be taken as zero. Moreover, the changes in kinetic energy and potential energy are also negligible. This leads to the expression

$$h_i = h_e \qquad (6.7)$$

Enthalpy being constant does not mean that the states of the flow before and after the throttle valve are exactly the same. In fact, as shown in Fig. 6.6, the fluid rapidly expands across the valve without having contributed any useful work.

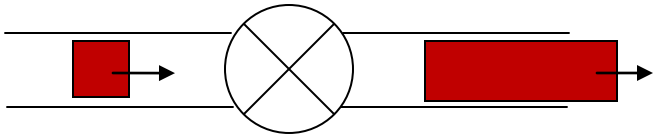


Fig. 6.6 Flow through a Throttle Valve

This type of rapid flow expansion across a valve is known as a 'throttling process' and since enthalpy is constant, it is also called as an 'isenthalpic process'. So far, we have discussed applications in which steady flow assumption can be applied to devices that are adiabatic (heat transfer is zero). We shall now look at steady flow devices which involve heat transfer.

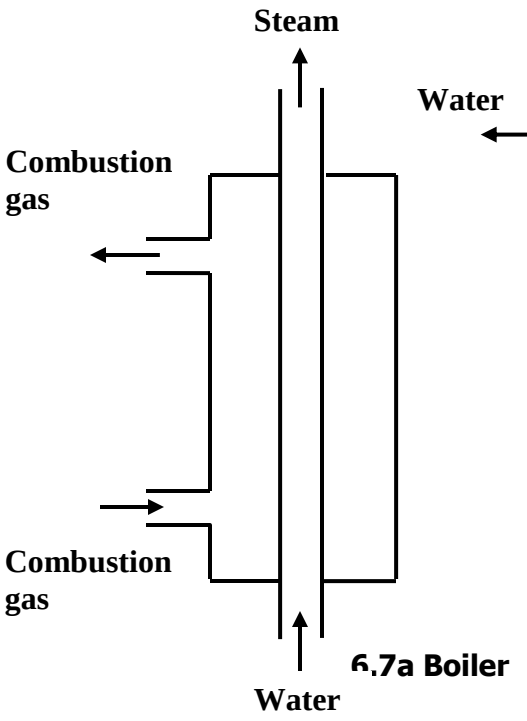
(ii) Heat exchange devices:

Devices such as steam generator (boiler), condenser , evaporator in a refrigerator etc. are examples of heat exchange devices. In a steam generator, water enters in liquid phase and due to heat addition, it is converted into steam at high temperature. No external work is done in the device and the changes in kinetic energy & potential energy are usually negligible. Therefore, the I law for steam generator can be simplified to the form

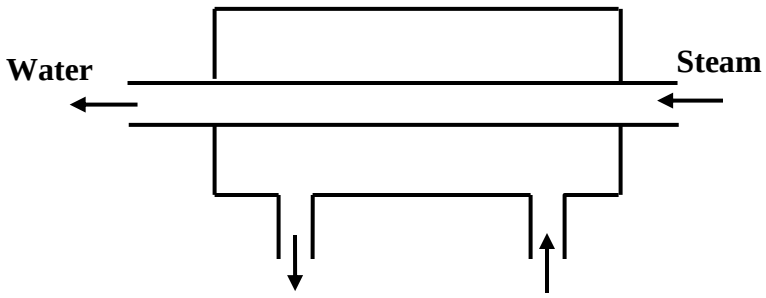
$$\dot{Q} = \dot{m}(h_e - h_i) \qquad (6.8)$$

The above form of I law is applicable to the condenser also. In a condenser, as seen from Fig. 6.2a, steam at low p and T is condensed back as water, by the removal of heat. Thus, Q > 0 for boiler while Q < 0 for condenser. Correspondingly, the enthalpy increases across the boiler while it decreases across the condenser. In a refrigerator, the refrigerant fluid evaporates by absorbing heat from food stuff, in the evaporator. Thus, Eq. (6.8) can be applied for the evaporator also, with a positive value of Q.

In devices like boiler or condenser, the heat addition or heat removal is not done directly. Usually, another fluid is used for this purpose. For example, in the case of boiler, the combustion gases generated from the burning of coal in air are circulated around tubes carrying water. Therefore, heat is transferred from the hot combustion gas to the water, for producing steam. Also, in the case of the condenser, sea water may circulated around tubes carrying steam for cooling purposes. There is no direct heat transfer between the surroundings and the CV in such cases. Thus, we can analyze these situations as devices with multiple inlet and exit streams, as shown in Figs. 6.7a and 6.7b.



6.7a Boiler



6.7b Condenser

The first law for these devices can be expressed as:

$$\sum_{e} \dot{m}_e h_e - \sum_{i} \dot{m}_i h_i = 0 \qquad (6.9)$$

between the different inlet and exit streams. In such an analysis, the heat transfer with the surrounding has been assumed as zero and the kinetic and potential energy changes are neglected.

Chapter 7 II Law of Thermodynamics

7.1 Introduction

The industrialized society today is anchored around automation, where manual/ animal labor is replaced by machine work. During industrial revolution, many machines were developed which were operated with high pressure steam. The steam, in turn, was produced by burning a fossil fuel such as coal. In course of time, engines using other fuels and power plants converting heat (derived from the burning of fossil fuels) into the easily transportable form of electrical power, came into existence. This has led to the wide spread development of technology in various areas, contributing immensely to improvement in the quality of life. Today, the quantity of electrical power produced is treated as a measure of the economic progress of any nation.

In Chapter 6, the general working principle of a thermal power plant was described- involving devices such as the steam generator, turbine, condenser and pump (Fig. 7.1). Water which is employed as the working fluid in power plant undergoes a cyclic process, with no permanent changes in its properties. The overall process occurring in a power plant, can therefore, be thought of as a conversion from thermal energy to useful work.

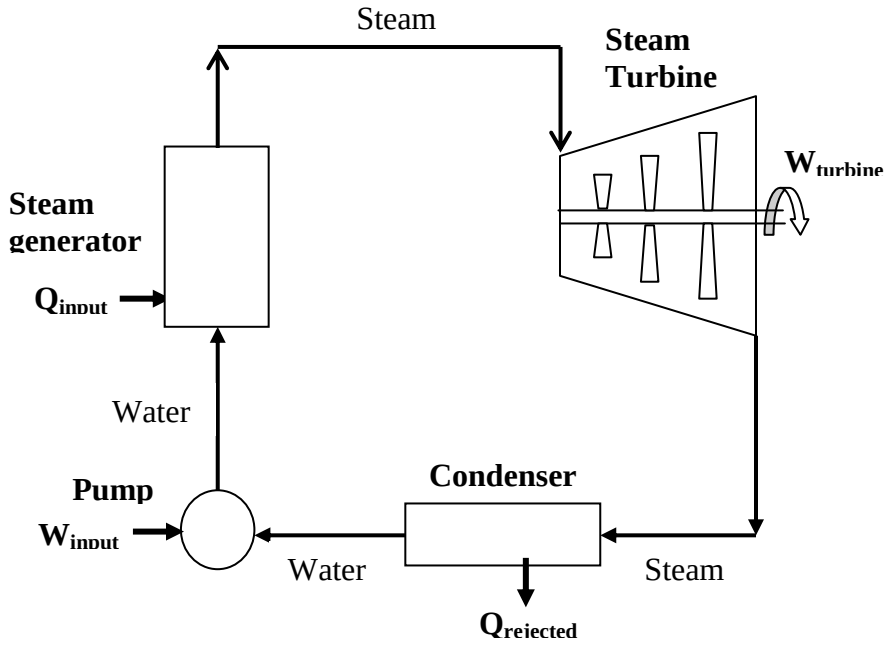


Fig. 7.1 Schematic of Thermal Power Plant

Apart from the water-steam based thermal power plant, one can also consider a closed cycle gas turbine power plant which could use air as the working fluid, as shown in Fig. 7.2. Air is pressurized in a compressor, it is heated to high temperature in a heater, then expanded in a turbine to produce work, and finally cooled back to the initial condition. This is also a cyclic process in which the working fluid does not undergo any permanent property change- but the overall process involves the conversion of heat into work.

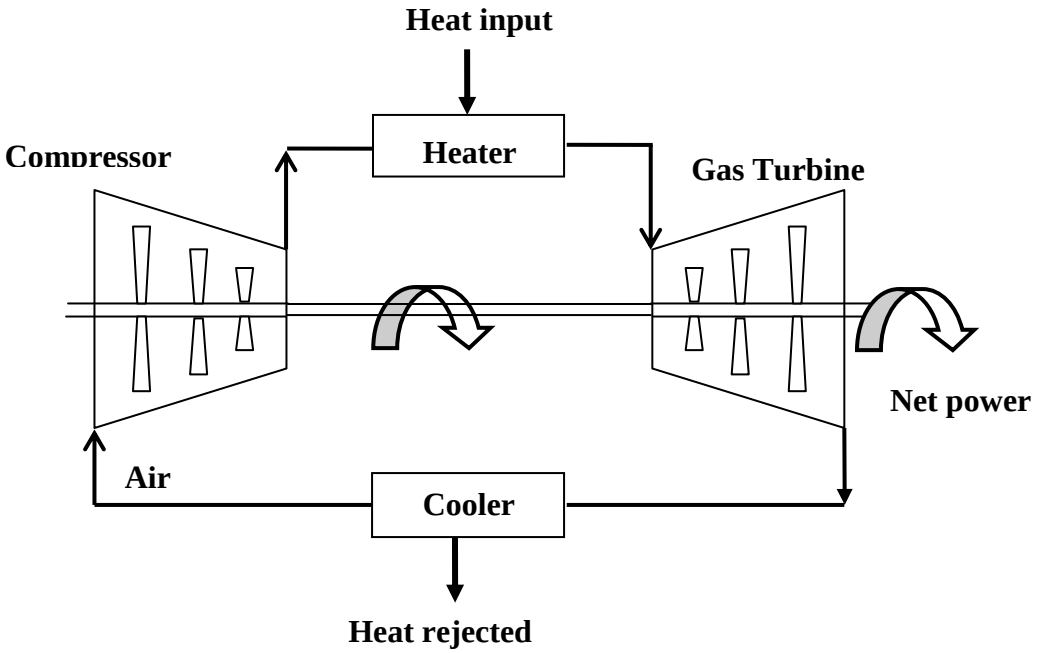


Fig. 7.2 Closed cycle gas turbine power plant

The above-described systems operate on closed cycles. There are also other systems such as the automotive engines based on gasoline and diesel fuels, aircraft engines etc. which operate on "open cycles'- involving heat rejection in the open atmosphere (by the release of exhaust gas into the atmosphere). All these power generation systems could be viewed as exchanging heat with a source and a sink, and producing net positive work.

7.2 Heat Engines, Heat Pumps and Refrigerators & their performance indices

We now define the concept of a 'Heat Engine'- as a system which exchanges heat with a source and a sink and produces positive work, while operating on a cyclic process. The source and sink are visualized as large thermal reservoirs- that is, a finite amount of

heat addition or heat removal will not alter the temperatures of these reservoirs. The source is at high temperature and it provides the heat input. The heat sink, on the other hand, is a system which receives the rejected heat from the heat engine and this is typically the environment that we live in. The temperature of the heat sink will be lower than the minimum operational temperature of the working fluid so that the waste heat can be rejected to the sink.

As shown in Fig. 7.3, 'H.E.' denotes the heat engine under consideration. Let Q_H be the heat input to the heat engine from the high temperature source and let Q_C be the heat rejected to the low temperature heat sink from the heat engine. Also, W_{net} represents the net positive work output of the heat engine. The dotted loop shown inside the heat engine implies that the working fluid within the heat engine operates on a cyclic process. During a part of the cycle it receives heat from the source and rejects some heat to the sink during another part of the cycle. As per the definitions given here, the steam power plant and the closed cycle gas turbine power plant will qualify to be called as heat engines. In a thermal power plant, the heat source will be the hot gas derived from the burning of coal and the heat sink will be the environment.

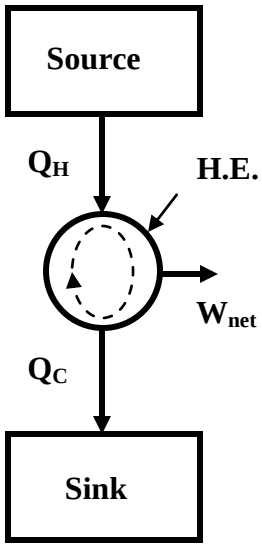


Fig. 7.3 Heat Engine

Since the heat engine operates on a cyclic process, there is no net energy or mass accumulation within the heat engine. Therefore, applying I law to the heat engine (which is a system), we get:

$$Q_H - Q_C = W_{net} \qquad (7.1)$$

The thermal efficiency of a heat engine can be defined as $\eta_{th} = \frac{W_{net}}{Q_H} = 1 - \frac{Q_C}{Q_H}$ (7.2)

For example, if a heat engine has a heat input of 100 kJ and it produces a work output of 60 kJ (while rejecting the remaining 40 kJ as waste heat to the sink), its thermal

efficiency η_{th} is equal to 60%. Only if the heat rejection is zero (i.e. all the heat input supplied to the heat engine is converted into work), the thermal efficiency will become 100%. However, as we shall discuss soon, the second law of thermodynamics states that such a scenario is impossible.

A heat engine is a system which converts a portion of the heat received from a heat source into useful work. The efficiency of the engine was defined in terms of the fraction of the heat input that is converted into work. Let us now turn our attention to a different class of systems which are employed in the pumping of heat from a low temperature level (T_C) to a high temperature level (T_H) as shown in Fig. 7.4. Schematically, the heat pump can be shown as a device with all the heat and work interactions in the opposite sense to those of the heat engine discussed earlier. Just as a water pump delivers water from a lower elevation to a higher elevation, the heat pump picks up at a lower temperature and delivers it at a higher temperature. Two cases are of interest in this category of devices- for instance: (i) a situation when the heat removed at low temperature (Q_C) is of interest to us; such devices are called "refrigerators" (ii) a situation when the heat delivered (Q_H) at high temperature is of interest to us; these devices are referred to as "heat pumps" only. Although the term 'heat pump' should be applicable to both the cases, the common usage corresponds to the specific situation when heat delivered at T_H is the quantity of interest.

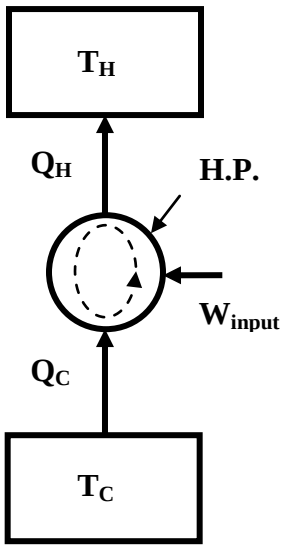


Fig. 7.4 Heat Pump

The performance of a heat pump or refrigerator is defined in terms of a parameter called as "COP" or "Coefficient Of Performance". For a refrigerator, the coefficient of performance is defined as

$$COP_{Ref} = \frac{Q_C}{W_{input}} \qquad (7.3\ a)$$

For a heat pump, the coefficient of performance is defined as

$$COP_{HP} = \frac{Q_H}{W_{input}} \qquad (7.3\ b)$$

Similar to a heat engine, the heat pump or refrigerator also operates on a cyclic process and therefore, there can no net energy accumulation or depletion. Hence,

$$Q_H = Q_C + W_{input} \qquad (7.4)$$

It is evident therefore that for the same values of Q_H, Q_C etc., $COP_{HP} = 1 + COP_{Ref}$. An important point to be kept in mind with reference to the above definitions is that here, Q_H, Q_C, W_{net}, W_{input} are all treated as positive quantities (i.e. only their magnitudes are considered without applying the usual sign conventions for heat and work). It is seen from Eqs. (7.3 a) and (7.3 b) that higher levels of performance imply higher values of COP for the refrigerator or the heat pump. A house owner would want his refrigerator to consume negligible electrical power i.e. $W_{input} \rightarrow 0$ or $COP_{Ref} \rightarrow \infty$ in order to cool the food articles to the desired low temperature level at a very low power cost. Similarly, in a room heating application, one may desire the heat pump to have infinite value of COP_{HP} (negligible power consumption). However, as we discussed in the case of a heat engine, the II law rules out such scenarios as impossible.

7.3 Statements of II Law of Thermodynamics

The II law of Thermodynamics can be stated in many equivalent forms. With reference to heat engines and heat pumps (or refrigerators), two forms of the II law are stated as follows:

The Kelvin- Planck statement of II law (for heat engines) :

"It is impossible to construct a heat engine which produces positive work by exchanging heat with a single thermal reservoir, while operating in a cycle".

Alternatively, "*It is impossible construct a heat engine with 100% thermal efficiency".*

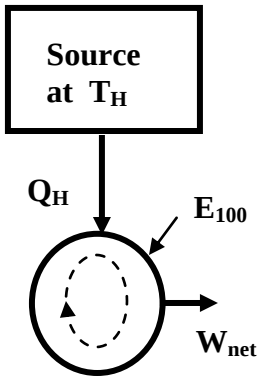


Fig. 7.5 Impossible Heat Engine with 100% efficiency

It may be pointed out here that the restriction applies only to engines operating on a cycle. For a once- through operation (non-cyclic process), 100% conversion of heat to work is possible. For example, consider an ideal gas producing work through isothermal expansion in a piston- cylinder device. For this non-cyclic process, Q = W (since $\Delta U = 0$ for the isothermal process of an ideal gas). Or, heat added to the system is equal to the work delivered during isothermal gas expansion. In non-cyclic processes such as this, material undergoes some property change such as increase in volume. Therefore, we cannot perpetually operate such processes, because the volume will become infinitely large to handle. On the other hand, in a cyclic process where material does not undergo any permanent change, the same process can be repeated again and again. Thus, when water undergoes cyclic changes (liquid → vapor → liquid) in a power plant, the power plant can be operated for an indefinite amount of time, so long as there is a high temperature source available for providing the input heat and a sink available for receiving the rejected heat.

Let us look into the thermal power plant operation a little deeply. After expansion of the steam in the turbine for production of work, do we really need a condenser where heat is rejected? What will happen if we dispense with the condenser, compress the steam back (preferably by an adiabatic process) and send it back to the steam generator? There are two problems with this procedure (i) The work involved in compression is very less if it happens in liquid phase, with the help of a pump because of the small liquid volume. If we try to compress vapor, the amount work involved will be enormous and almost the entire turbine work may get consumed in compression (ii) Compression alone cannot bring the working fluid back to its initial state for carrying out the cyclic process. Without heat rejection in the condenser, the working fluid will undergo an unclosed process, with increase in volume for ever (see figure below).

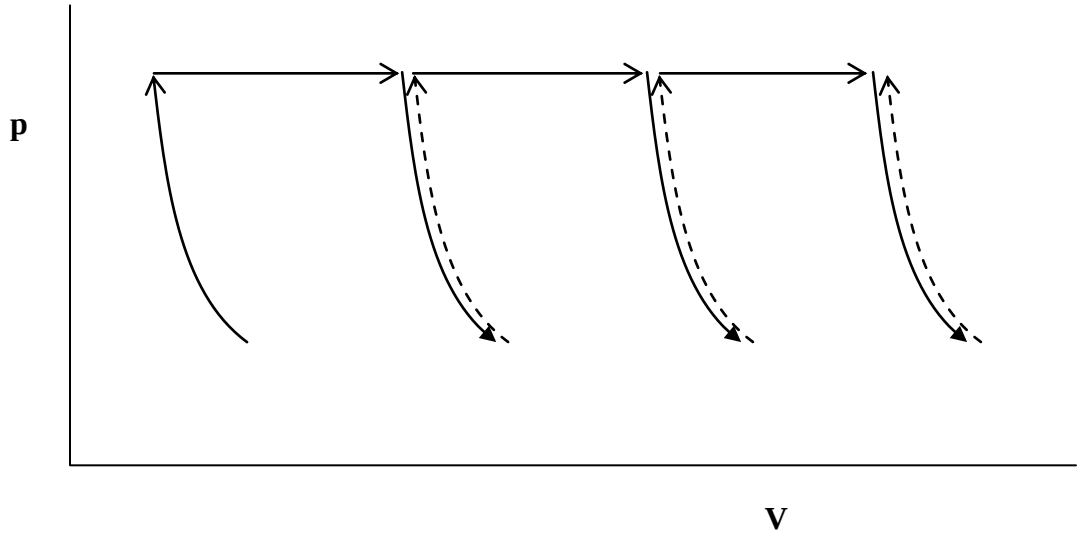


Fig. 7.6 Power production without heat rejection

The Clausius statement of II law (for heat pumps/ refrigerators):

"*It is impossible to construct a heat pump or refrigerator which can pump heat from a low temperature reservoir to a high temperature reservoir without any work input, while operating in a cycle*"

Alternatively, *"It is impossible to construct a heat pump or refrigerator which has infinite COP".*

Schematically, the Clausius statement of second law can be illustrated as shown below. This will correspond to a machine with $Q_H = Q_C$ and $W_{input} = 0$.

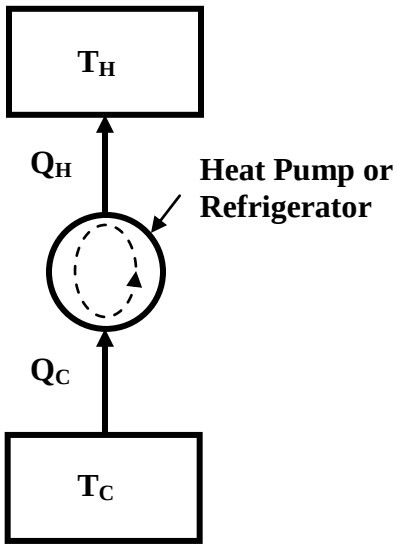


Fig. 7.7 Impossible Heat Pump or Refrigerator with infinite COP

Although the two statements of II law credited to Kelvin- Planck (KP) and Clausius appear to be vastly different, they are actually equivalent- in the sense that violation of one will lead to the automatic violation of the other. In other words, if we assume that a 100% efficient heat engine exists, we will end up proving that an infinite COP heat pump or refrigerator also exists. Similarly, violation of the Clausius statement will lead to automatic violation of the KP statement also.

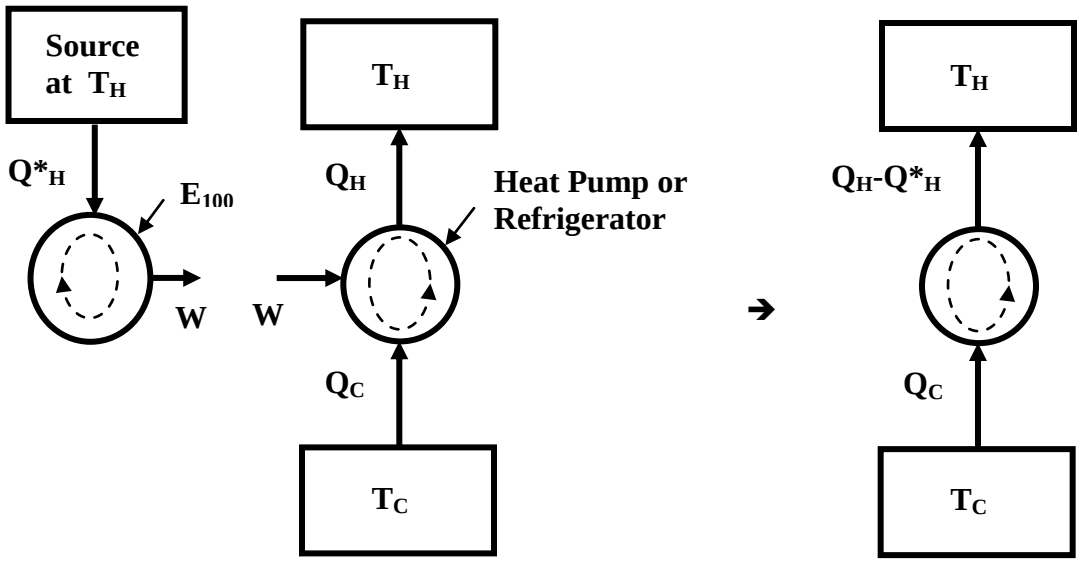


Fig. 7.8 Violation of KP statement leads to violation of Clausius statement

Figure 7.8 clearly shows that if 100% efficient heat engine exists, it can be combined with a normal heat pump (or refrigerator) to produce an infinite COP heat pump or refrigerator. Having shown that KP statement of II law and Clausius statement of II law are equivalent, we now consider the feasible limits on the performance parameters of the heat engine and heat pump (or refrigerator) in the next section.

7.4 Reversible Heat Engines and Reversible Heat Pumps/ Refrigerators

The natural questions that one may ask are: If we cannot achieve 100% thermal efficiency for a heat engine, then what is the maximum that we can achieve? If infinite COP is not possible, what is the maximum COP that can be achieved? The answers to these questions take us to the definition of a new concept- namely, the concept of Reversible Heat Engines and Reversible Heat Pumps (or Refrigerators). Most of the devices that we know cannot perform reversed functions. For example, an automobile can move forward by burning fuel with air and disperse the exhaust (CO_2, H_2O, N_2 etc.) into the atmosphere. Suppose we were to drag the same automobile in the backward direction, it will not absorb CO_2, H_2O and N_2 from atmosphere and produce fuel and air! However, in some cases reversing may be possible, albeit at a lower efficiency. For example consider the combination of water pump driven by an electrical motor which draws current input (see Fig. 7.9 a). If we reversed the direction of water flow, it is possible to run the pump as a turbine and the motor as an electrical generator, so that water falling with a certain velocity can produce electrical power (7.9 b). However, a device that works efficiently as a pump will have extremely poor efficiency as a turbine.

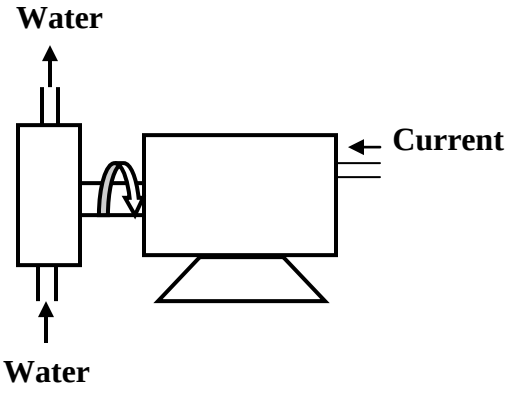


Fig. 7.9 a Motor & Pump

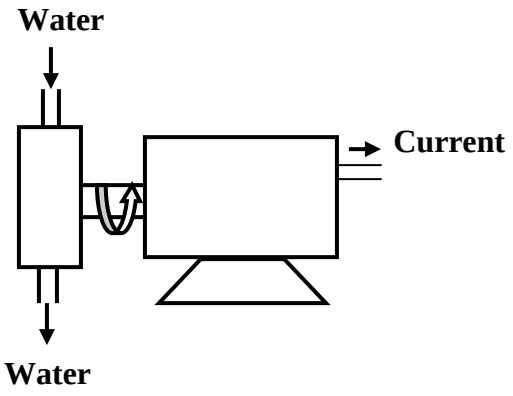


Fig. 7.9 b Turbine & Generator

A Reversible Heat Engine is a device that works with the same level of performance- as a heat engine or as a heat pump. Consider a reversible heat engine which takes 100 kJ heat input (Q_H) at the source temperature of T_H = 1000 K and delivers a work output of W_{net} = 70 kJ and rejects the heat of 30 kJ (Q_C) at the sink temperature of T_C = 300 K. By definition, this heat engine can be reversed in its operation into a heat pump, with the heat input of 30 kJ (Q_C) at 300 K and heat rejection (Q_H) of 100 kJ at 1000 K, and a work input of 70 kJ. These two scenarios are shown schematically in Figs. 7.10a and 7.10b, respectively.

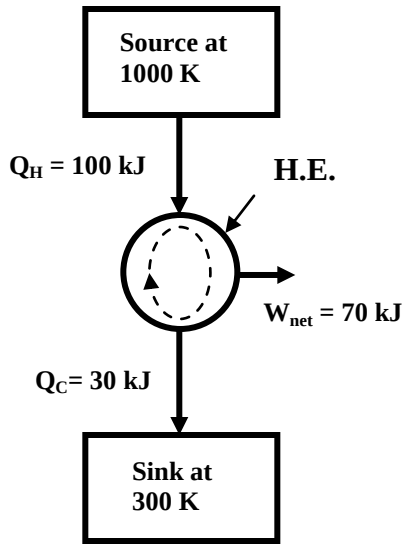


Fig. 7.10a Heat Engine

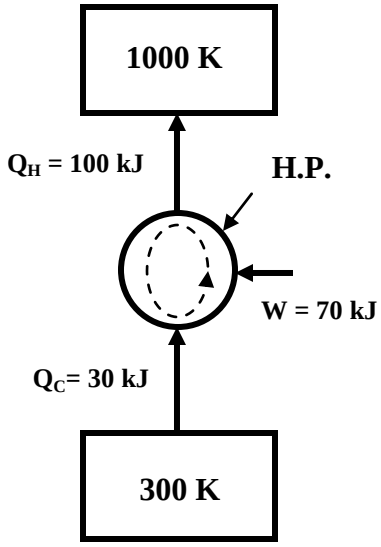


Fig. 7.10b Heat Pump

The criterion for a heat engine to be termed as a reversible heat engine is as follows:

$$\eta_{th} = \frac{W}{Q_H} = \frac{1}{COP_{HP}} \qquad (7.5)$$

In other words, if the thermal efficiency of heat engine is equal to the reciprocal of COP when the device is operated as a heat pump, such a device would be a Reversible Heat Engine. It is evident that a Reversible Heat Engine is also a Reversible Heat Pump (or Reversible Refrigerator). The main criterion is that the same values of Q_H and Q_C should be possible between the same source and sink temperatures, for both modes of operation.

Reversible heat engines and reversible heat pumps (& refrigerators) have many interesting properties, which are listed below.

(i) For given values of T_H and T_C, the maximum thermal efficiency can be attained only by a reversible heat engine. Similarly, the maximum COP as a heat pump (or as a refrigerator) can also be attained only by a reversible heat pump (or reversible refrigerator).

(ii) All reversible heat engines operating between the same T_H and T_C, have the same thermal efficiency, irrespective of the working fluid or material of construction for the device. Similarly all reversible heat pumps (or reversible refrigerators) operating between the same T_H and T_C, have the same COP, irrespective of the working fluid or material of construction.

(iii) The thermal efficiency of a reversible heat engine, COP of a reversible heat pump and COP of a reversible refrigerator are dependent only on the temperature limits T_H and T_C .

These statements can be proved as the corollaries of II law. It is important to note here that the II law of thermodynamics itself (in any one of its forms) has to be treated as a law of nature, derived from physical observations. Based on II law, rigorous proofs can be provided for each of the statements listed above.

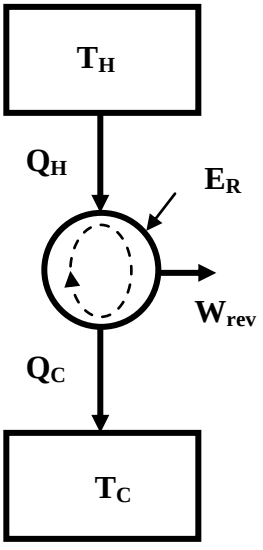


Fig. 7.11a Reversible Heat Engine E_R

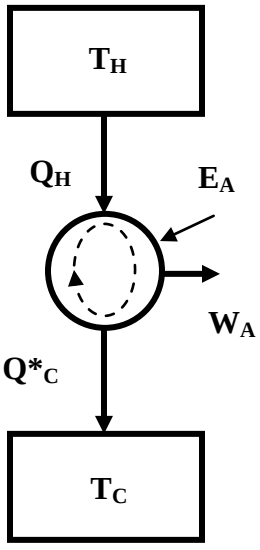


Fig. 7.11b Irreversible Heat Engine E_A

Let us consider the two heat engines shown above. Engine E_R in Fig. 7.11a is a reversible heat engine whose work output for the heat input of Q_H is W_{rev}. Engine E_A in Fig. 7.11b is an irreversible heat engine whose work output for the same heat input of Q_H is W_A. Let us for a moment assume that the irreversible engine E_A has higher efficiency than the reversible engine E_R. Since the heat input is the same (= Q_H), this implies that $W_A > W_{rev.}$ Let us now operate the reversible engine as a heat pump (since it can operate both ways) and connect the work output of engine E_A to the work input of heat pump. This results in the following scenario.

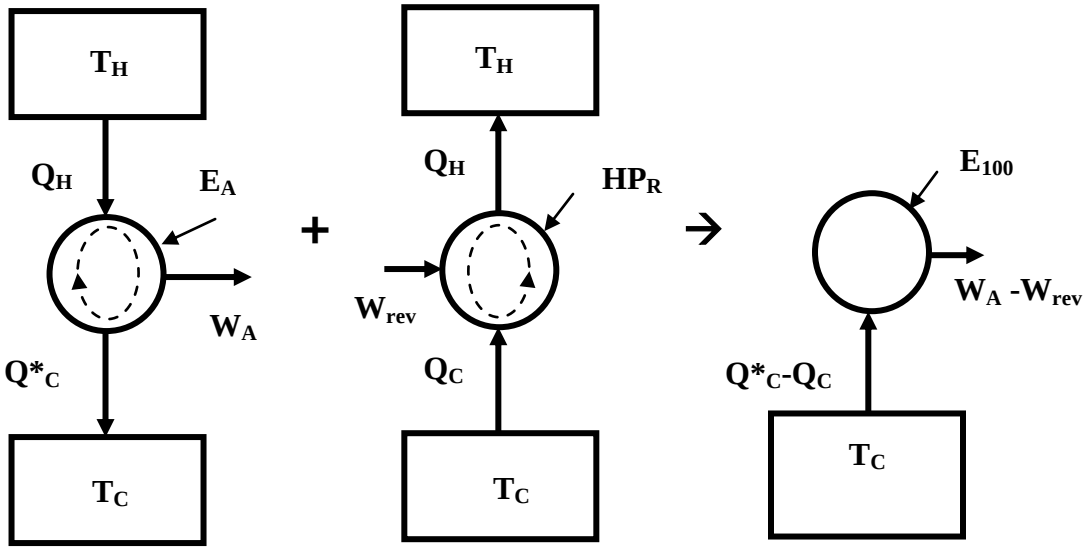


Figure 7.12

For the combined system of the heat engine E_A and the reversible heat pump HP_R, the heat source at T_H involves no net heat transfer (Q_H-Q_H=0) and this is equivalent to not connecting to the reservoir at T_H. Thus, the assumption that the irreversible heat engine E_A is more efficient than the reversible heat engine E_R results in the violation of the II law (positive work output is produced from heat transfer with a single thermal reservoir). In a similar manner, each property of reversible heat engines/ heat pumps/ refrigerators can be proved.

7.5 Absolute Temperature Scale

The property (iii) described above implies that the thermal efficiency of a reversible heat engine can be expressed as $\eta_{th} = f(T_H, T_C)$ only. Since by definition thermal efficiency is given as

$$\eta_{th} = 1 - \frac{Q_C}{Q_H}$$

it is clear that there must be some relationship between Q_H, Q_C, T_H and T_C. It is possible to think of a new temperature scale in which the heat transfer and the corresponding temperature bear the relationship

$$\frac{Q_H}{T_H} = \frac{Q_C}{T_C} \tag{7.6}$$

This implies that the thermal efficiency of a reversible heat engine is given as

$$\eta_{th} = 1 - \frac{Q_C}{Q_H} = 1 - \frac{T_C}{T_H} \tag{7.7}$$

in this temperature scale. In our earlier discussions on temperature measurement, it was shown that temperature can be measured using any property that depends on temperature, namely: the length of an object, ideal gas law (pV = mRT), voltage difference of thermocouples, resistance variation of a metallic wire, etc. Here, we use the fact that the thermal efficiency of a reversible heat engine is only a function of the temperature limits, to define a new temperature scale. Indeed this temperature scale is the same as the Kelvin scale that was already discussed in connection with the ideal gas behavior.

Here we provide a limited proof of Eq. (7.6) considering the working fluid as an ideal gas. As per the descriptions given so far, the reversible heat engine takes heat input (Q_H) at a constant source temperature T_H and rejects heat (Q_C) at a constant sink temperature T_C. Note that adiabatic processes (as in pump & turbine for the thermal

power plant) can exist in between the isothermal heat transfer processes, which complete the cycle. Let us consider four processes as listed below.

(i) Isothermal heat addition process 1-2, with heat input Q_H at T_H

(ii) Adiabatic expansion process 2-3 with work output W_{2-3}

(iii) Isothermal heat rejection process 3-4, with heat removal Q_C at T_C

(iv) Adiabatic compression process 4-1 with work input W_{4-1}

For the isothermal process 1-2 (since $\Delta U = 0$), $Q_H = Q_{1-2} = W_{1-2} = p_1V_1 \ln(V_2/V_1)$.

For the isothermal process 3-4, similarly $Q_C = Q_{3-4} = W_{3-4} = p_3V_3 \ln(V_3/V_4)$, keeping in mind that Q_C represents only the magnitude of the heat rejected (without the sign).

Thermal efficiency $\eta_{th} = 1 - \dfrac{Q_C}{Q_H} = 1 - \dfrac{p_3V_3 \ln\left(\dfrac{V_3}{V_4}\right)}{p_1V_1 \ln\left(\dfrac{V_2}{V_1}\right)} = 1 - \dfrac{mRT_3 \ln\left(\dfrac{V_3}{V_4}\right)}{mRT_1 \ln\left(\dfrac{V_2}{V_1}\right)}$.

For the adiabatic processes, $pV^{\gamma} = \text{constant}$ implies that $T.V^{\gamma-1} = \text{constant}$. Therefore,

$$\frac{T_2V_2^{\gamma-1}}{T_3V_3^{\gamma-1}} = 1 \ for \ process \ 1-2; similarly, \frac{T_1V_1^{\gamma-1}}{T_4V_4^{\gamma-1}} = 1 \ for \ process \ 4-1.$$

Since $T_1 = T_2$ and $T_3 = T_4$, the above expression simplifies to

$\dfrac{V_2}{V_3} = \dfrac{V_1}{V_4}. Or, \dfrac{V_2}{V_1} = \dfrac{V_3}{V_4}$. Using this result in the expression for the thermal efficiency gives:

$$\eta_{th} = 1 - \frac{Q_C}{Q_H} = 1 - \frac{mRT_3 \ln\left(\dfrac{V_3}{V_4}\right)}{mRT_1 \ln\left(\dfrac{V_2}{V_1}\right)} = 1 - \frac{T_3}{T_1} = 1 - \frac{T_C}{T_H}.$$

This shows that $(Q_H/T_H) = (Q_C/T_C)$ for a reversible heat engine. This result is true for a reversible heat pump or reversible refrigerator also.

Now, the thermal efficiency of a reversible heat engine $\eta_{th} = 1 - (T_C/T_H)$.

The COP of a reversible heat pump is $COP_{HP} = T_H/(T_H-T_C)$.

The COP of a reversible refrigerator is $COP_{ref} = T_C/(T_H-T_C)$.

For instance, the maximum thermal efficiency achievable between the temperatures of 1000 K and 300 K is equal to: (1- 300/1000) x 100% = 70%. Between the same temperatures, the maximum achievable COP of the heat pump = 1000/700 = 1.4286. The maximum achievable COP of Refrigerator = 300/700 = 0.4286. These can be seen from the reversible device configurations shown in Figs. 7.10a and 7.10b.

7.6 Reversible and Irreversible Processes

The typical cycle undergone by a reversible heat engine is shown in Fig. 7.13a. For the engine to be reversible, the cycle must be a reversible cycle and for the cycle to be reversible, each process must be reversible. Thus, the reversible cycle 1-2-3-4-1 shown in Fig. 7.13a can be stated precisely as

1-2: Reversible isothermal heat addition (at T_H)

2-3: Reversible adiabatic expansion

3-4: Reversible isothermal heat rejection (at T_C)

4-1: Reversible adiabatic compression

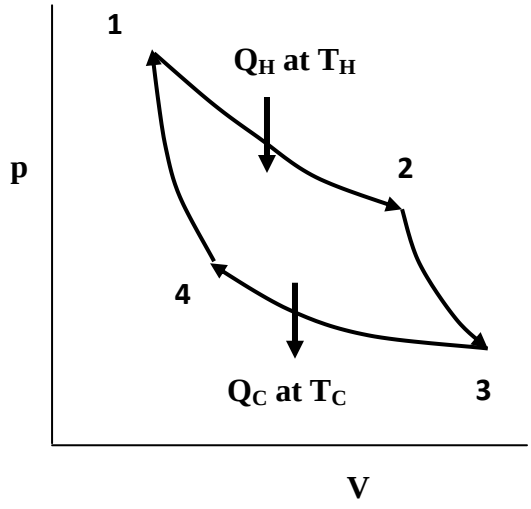


Fig. 7.13a Carnot Heat Engine Cycle

The p-V diagram shown here corresponds to a reversible heat engine cycle. Such an engine is known by the name of Carnot Engine and the corresponding cycle is called as

the Carnot cycle. Now, for the same value of Q_C, Q_H, T_C and T_H, the corresponding heat pump (or refrigerator) cycle will have a similar form as that in Fig. 7.13a, except that all the arrows will be pointing in the opposite direction. The Carnot Heat Pump or Carnot Refrigerator cycle is shown in Fig. 7.13b. Please note that the cycle is the same for a heat pump or refrigerator; it is only the desired heat transfer which is different between the two systems.

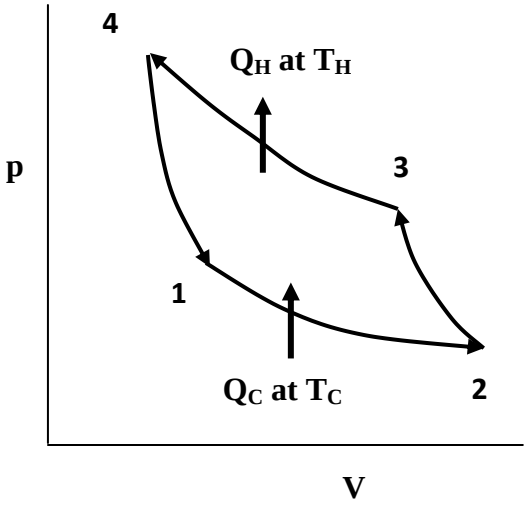


Fig. 7.13b Carnot Heat Pump/ Refrigerator Cycle

The processes can be defined as

1-2: Reversible isothermal heat addition (at T_C)

2-3: Reversible adiabatic compression

3-4: Reversible isothermal heat rejection (at T_H)

4-1: Reversible adiabatic expansion

It is clear that reversible heat engine or reversible heat pump/refrigerator is based on a reversible cycle. In a reversible cycle each process is reversible. Now, what is reversible process? How do we define a reversible process?

There are two important criteria for a process to be termed as a "reversible process".

(i) The process must be very slow (quasi-static), such that at each instant the system passes through an equilibrium state. There should not be any non-equilibrium effects.

(ii) No dissipative factors such as friction, electrical resistance, viscosity, etc. should be present.

Thus, a fully resisted slow expansion of a gas in a frictionless adiabatic piston- cylinder device is a reversible expansion process. Isothermal evaporation of water into steam by slow heat addition is a reversible process. On the other hand, heating of water in a vessel at atmospheric pressure with the help of a flame is an irreversible process, because of the large ΔT between the flame ($\sim 1800^{o}C$) and the water (less than $100^{o}C$). Large ΔT implies lack of thermal equilibrium. Rapid expansion of a gas when $p_{sys} >> p_{surr}$ is irreversible due to lack of mechanical equilibrium. Heating of a resistor by passage of current is irreversible (dissipative process). Fuel combustion is an irreversible process due to lack of chemical equilibrium. The various irreversibilities that are commonly encountered are listed below.

a) Irreversibilities due to lack of equilibrium: unresisted expansion, heat transfer due to finite ΔT, species diffusion because of concentration gradient, spontaneous (fast) reactions
b) Irreversibilities due to dissipative effects: solid friction, viscosity, ohmic resistance, magnetic hysteresis, plastic deformation

Irreversibilities can be classified as internal irreversibilities or external irreversibilities depending on whether it occurs inside or outside the system. If water boils at $100^{o}C$ when it is heated by a flame at 1 atmosphere pressure, the boiling process can be treated as internally reversible. (In other words, lack of thermal equilibrium occurs outside the water which is considered as the system). When a gas is throttled by a flow control valve, the throttling process is irreversible and the irreversibility in this case is internal (In fact, due to the irreversibility, even though the gas expands in volume, no useful work is delivered during the throttling process).

An irreversible heat engine operates on an irreversible cycle and at least one process in the cycle may be irreversible. Irreversibilities always reduce the amount of work that can be derived from the working substance. Therefore, for the same heat input, the work delivered by an irreversible engine is less than the work delivered by the reversible engine under the same temperature limits. Consequently, the thermal efficiency of the irreversible heat engine is less than that of reversible engine for the same values of T_H and T_C.

Chapter 9 Practical cycles

9.1 Introduction

In Chapters 7 and 8, it was shown that a reversible engine based on the Carnot cycle (two reversible isothermal heat transfers and two reversible adiabatic work processes) provides the highest level of conversion from heat to work, for given maximum and minimum temperature limits. In common practice, we often find that when heat is added to a substance, its temperature starts increasing. Only during a phase change process, it is possible to have isothermal heat transfer to the working substance. Therefore, one may consider the option of carrying out the entire cycle within the saturated liquid- vapor mixture regime as shown in Fig. 9.1.

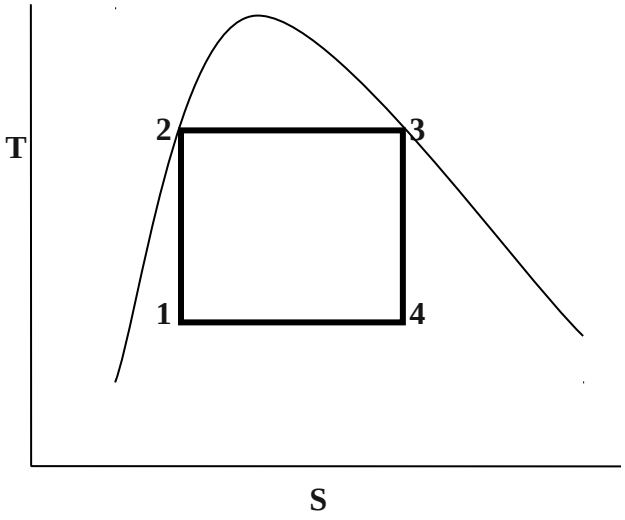


Fig. 9.1 Carnot engine cycle with phase change substance

Here, process 2-3 corresponds to the boiling of saturated liquid into saturated vapor by reversible heat addition in a boiler; process 3-4 represents the expansion of the saturated vapor reversibly and adiabatically in a turbine; process 4-1 represents the partial condensation of vapor into liquid by isothermal heat rejection in a condenser; process 1-2 represents the adiabatic compression of the mixture in a pump. Although theoretically the above Carnot cycle seems feasible, there are some practical difficulties in carrying out the cycle completely within the saturated liquid- vapor dome.

When a pump is operated in the liquid –vapor mixture regime, it suffers from the problem of cavitation which severely affects the performance and operational life of the pump. Therefore, it is preferable that the adiabatic compression process 1-2 be shifted to the left (outside the mixture regime) into the sub-cooled liquid region. Furthermore, when the pump operated with liquid only, the work input needed will be small because of the small liquid volume. In the case of the turbine, the turbine blades will be severely eroded when a large number of liquid droplets impinge on their surface at high speed. Therefore for long operational life of the turbine blades, it is essential to shift the adiabatic expansion process 3-4 to the right, completely into the superheated vapor regime. In any case, even if the vapor at the turbine exit is slightly wet (with a few droplets), the mixture quality should be very close to one.

Now, with state 2 lying in the sub-cooled regime and state 3 lying in the superheated vapor regime, it is not possible to have isothermal heat addition in the boiler. In fact, process 2-3 is better approximated as constant pressure heat addition, since boiler is essentially a constant pressure device. Similarly, the condenser is also a constant pressure device; however, unlike in the case of boiler, the non-isothermal heat transfer part in a practical condenser is almost negligible. Replacing the isothermal heat addition and heat rejection processes of Carnot cycle by corresponding constant pressure heat addition and constant pressure heat rejection, results in a new cycle known as the **Rankine cycle** for steam power plants.

9.2 Rankine Cycle

The simplest form of Rankine cycle that could be used for converting from heat to work, is made up of the following processes:

(i) Reversible adiabatic compression process 1-2 (in Pump)

(ii) Reversible constant pressure heat addition process 2-3 (in Boiler)

(iii) Reversible adiabatic expansion process 3-4 (in Steam Turbine)

(iv) Reversible constant pressure heat rejection process 4-1 (in Condenser)

A schematic diagram indicating the various devices of this power plant cycle was shown in Fig. 7.1. Water- steam substance is employed as the working fluid in this cycle. The basic Rankine cycle in T-S coordinates is shown in Fig. 9.2. In the practical Rankine cycle of thermal power plants, additional complexities such as the reheating of steam after partial expansion in the turbine, regenerative heating of water before entry to the boiler with the help of some wet steam bled from turbine (after partial expansion), are also incorporated.

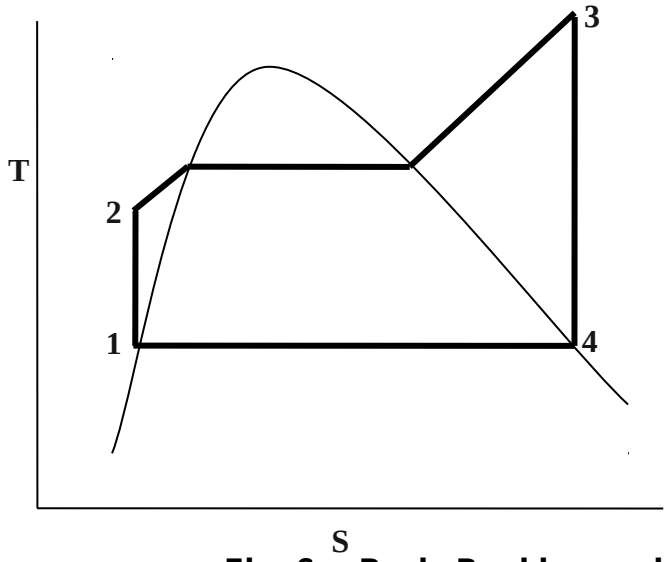


Fig. 9.2 Basic Rankine cycle

Neglecting KE and PE changes, the I law application to each device gives:

(i) Rate of heat input in boiler $\dot{Q}_H = \dot{m}\{h_3 - h_2\}$ (9.1)

(ii) Turbine power output $\dot{W}_T = \dot{m}\{h_3 - h_4\}$ (9.2)

(iii) Rate of heat rejection in condenser $\dot{Q}_C = \dot{m}\{h_1 - h_4\}$ (9.3)

(iv) Power input to pump $\dot{W}_P = \dot{m}\{h_1 - h_2\} \sim -\dot{m}\frac{\Delta p}{\rho}$ (9.4)

While evaluating the pump power, since ΔT is negligible across the pump, we can approximate $\Delta h \sim \Delta p/\rho$.

Net power = $\dot{W}_{net} = \dot{m}\{h_3 - h_4\} - \dot{m}\{h_2 - h_1\} = \dot{Q}_H - \dot{Q}_C = \dot{m}\{(h_3 - h_2) - (h_4 - h_1)\}$ (9.5)

The enthalpy values at different states of the water-steam substance can be obtained from the steam tables, for evaluating the heat and work interactions in the cycle.

9.2 Gas Turbine (Brayton) cycle

The Brayton cycle is based on the closed cycle gas turbine power plant shown in Fig. 7.2. It uses air as the working fluid. The simplest form of the Brayton cycle in T-S coordinates is shown in Fig. 9.3. Similar to the Rankine cycle, the industrial Brayton cycle may include additional processes such as reheating after partial expansion, regenerative heating with turbine exhaust etc.

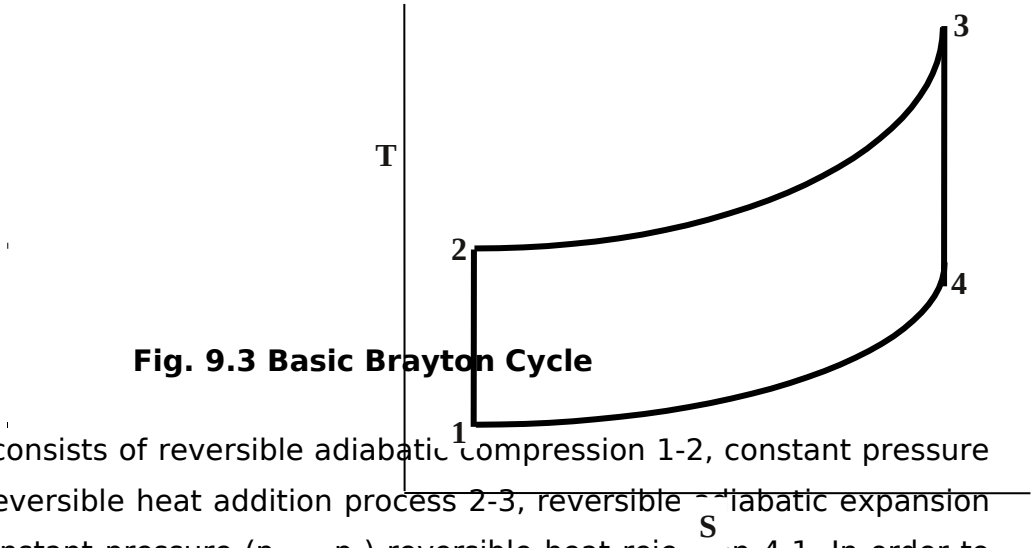


Fig. 9.3 Basic Brayton Cycle

The cycle consists of reversible adiabatic compression 1-2, constant pressure ($p_2 = p_3$) reversible heat addition process 2-3, reversible adiabatic expansion 3-4 and constant pressure ($p_4 = p_1$) reversible heat rejection 4-1. In order to find the properties of the working fluid (air), we can employ ideal gas relations. Thus, the heat and work interactions can be evaluated as:

(i) Power input to compressor $\dot{W}_{comp} = \dot{m}\{h_1 - h_2\} = \dot{m}C_P(T_1 - T_2)$ (9.6)

(ii) Rate of heat input in heater $\dot{Q}_H = \dot{m}\{h_3 - h_2\} = \dot{m}C_P(T_3 - T_2)$ (9.7)

(iii) Turbine power output $\dot{W}_T = \dot{m}\{h_3 - h_4\} = \dot{m}C_P(T_3 - T_4)$ (9.8)

(iv) Rate of heat rejection in cooler $\dot{Q}_C = \dot{m}\{h_1 - h_4\} = \dot{m}C_P(T_1 - T_4)$ (9.9)

In addition to the above relations, for the reversible adiabatic processes 1-2 and 3-4, the pressure ratios and temperature ratios are related as follows:

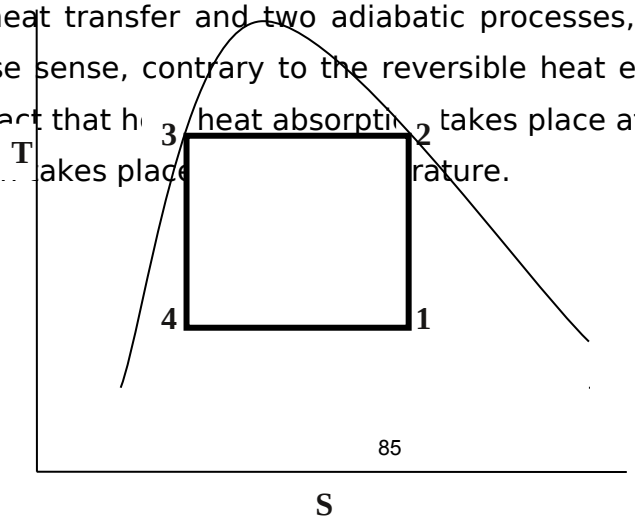
$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\frac{\gamma}{\gamma-1}} = \frac{p_3}{p_4} = \left(\frac{T_3}{T_4}\right)^{\frac{\gamma}{\gamma-1}} \qquad (9.10)$$

Thus, if the pressure ratio (p_2/p_1), compressor inlet temperature (T_1) and the turbine inlet temperature (T_3) are available, all the heat and work interactions can be calculated.

9.3 Vapor Compression Refrigeration cycle

A Carnot refrigeration (or heat pump) cycle based on a phase change substance as working fluid, is shown in Fig. 9.4. The working substances used in refrigerators are some refrigerant fluids with commercial names such as R-12, R-134a etc.

Although the reversible heat pump/ refrigerator cycle also consists of two isothermal heat transfer and two adiabatic processes, it is traversed in the anti-clockwise sense, contrary to the reversible heat engine cycle. This is in view of the fact that h[illegible] heat absorpti[illegible] takes place at low temperature and heat rejectio[illegible] takes place[illegible]rature.



9.4 Reversible Heat Pump/ Refrigerator cycle

The practical refrigerator cycle differs from the reversible Carnot cycle in two important ways. Firstly, the adiabatic compression process has to be shifted to the superheated vapor regime, since the compressor cannot handle liquid-vapor mixture. Additionally, the reversible expansion process 3-4 would require a frictionless adiabatic turbine or piston- cylinder device which are actually not necessary- considering the fact that the work that can be delivered by this process will be negligibly small (because of the low mass flow rate of the refrigerant and also the small volume of the predominantly liquid mixture). Therefore, instead of using a work-producing expansion device, in a practical refrigerator, a very cheap throttle valve can be used to carry out irreversible expansion. In fact, with the help of friction, a very large pressure drop is introduced in the refrigerant flow which results in flashing (rapid phase change from liquid to vapor phase because of sudden decrease in pressure). Due to flashing, the temperature of the mixture will reduce significantly, because the latent heat needed for evaporation is derived from the sensible heat of the refrigerant itself. The throttling process occurring in the throttle valve is an isenthalpic process, as explained in Chapter 6. Therefore, for the irreversible expansion process 3-4, $h_3 = h_4$. With the two changes described above, the basic cycle of a real refrigerator is shown in Fig. 9.5.

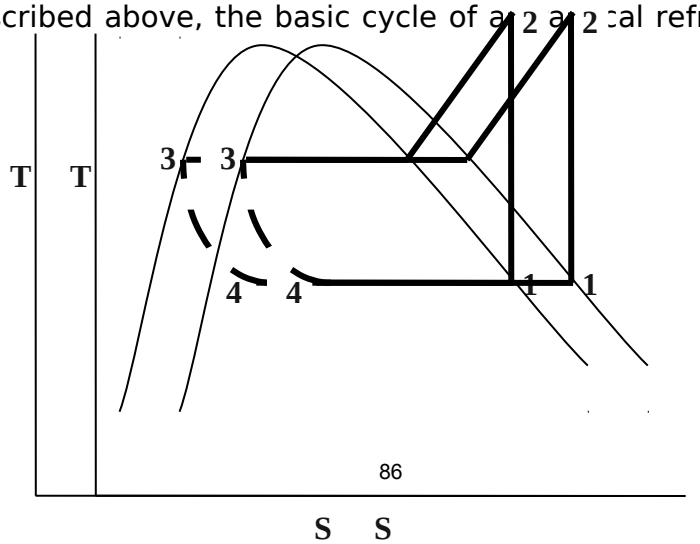


Fig. 9.5 Basic Vapor Compression Refrigeration Cycle

The cycle consists of the following processes.

(i) Reversible adiabatic compression of the refrigerant vapor 1-2
(ii) Reversible constant pressure heat rejection in condenser 2-3
(iii)Adiabatic expansion in throttle valve (irreversible) 3-4
(iv) Reversible constant pressure heat absorption in evaporator 4-1

The heat and work interactions can be evaluated as:

Power input to compressor $\dot{W}_{comp} = \dot{m}\{h_1 - h_2\}$ (9.11)

Rate of heat rejection in condenser $\dot{Q}_H = \dot{m}\{h_3 - h_2\}$ (9.12)

Rate of heat absorption in evaporator $\dot{Q}_C = \dot{m}\{h_1 - h_4\}$ (9.13)

For the throttle valve, $h_3 = h_4$. (9.14)

In all the three cycles presented above, although the ideal cycle may consider the adiabatic processes to be reversible (i.e. isentropic), in reality, there may be entropy production because of irreversibilities such as friction. The performance of the associated device can be expressed in terms of isentropic efficiency as discussed in Chapter 8.

Saturated Steam Temperature Table

T , ^{o}C	p, bar	v_f, m^3/kg	v_g, m^3/kg	u_f, kJ/kg	u_g, kJ/kg	h_f, kJ/kg	h_g, kJ/kg	s_f, kJ/kg	s_g, kJ/kg
0.01	0.0061	1.0002×10^{-3}	206.1	0.01	2376	0.01	2501	0.0	9.156
5.0	0.0087	1.0001×10^{-3}	147.1	21.00	2383	21.00	2511	0.0762	9.026
10.0	0.0123	1.0001×10^{-3}	106.4	42.01	2389	42.01	2520	0.151	8.901
15.0	0.0170	1.0007×10^{-3}	77.93	62.99	2396	62.99	2529	0.2245	8.781
20.0	0.0234	1.002×10^{-3}	57.79	83.94	2403	83.94	2538	0.2966	8.667
25.0	0.0317	1.0032×10^{-3}	43.36	104.9	2410	104.9	2547	0.3673	8.558
30.0	0.0425	1.0045×10^{-3}	32.90	125.8	2416	125.8	2556	0.4369	8.453
35.0	0.0563	1.0057×10^{-3}	25.22	146.7	2423	146.7	2565	0.5053	8.353
40.0	0.0738	1.0076×10^{-3}	19.52	167.5	2430	167.5	2574	0.5725	8.257
45.0	0.0959	1.010×10^{-3}	15.26	188.4	2437	188.4	2583	0.6386	8.165
50.0	0.1235	1.012×10^{-3}	12.03	209.3	2443	209.3	2592	0.7037	8.076
55.0	0.1576	1.015×10^{-3}	9.569	230.2	2450	230.2	2601	0.7679	7.991
60.0	0.1994	1.017×10^{-3}	7.671	251.1	2457	251.1	2610	0.8311	7.910
65.0	0.2503	1.020×10^{-3}	6.197	272.0	2463	272.0	2618	0.8934	7.831
70.0	0.3119	1.023×10^{-3}	5.042	293.0	2470	293.0	2627	0.9549	7.755
75.0	0.3858	1.026×10^{-3}	4.131	313.9	2476	313.9	2635	1.016	7.682
80.0	0.4739	1.029×10^{-3}	3.407	334.8	2482	334.9	2644	1.075	7.612
85.0	0.5783	1.033×10^{-3}	2.828	355.8	2488	355.9	2652	1.134	7.544
90.0	0.7013	1.036×10^{-3}	2.361	376.8	2494	376.9	2660	1.193	7.479
95.0	0.8455	1.039×10^{-3}	1.982	397.9	2501	398.0	2668	1.250	7.416
100.0	1.013	1.044×10^{-3}	1.673	418.9	2507	419.0	2676	1.307	7.355
110.0	1.433	1.052×10^{-3}	1.210	461.1	2518	461.3	2691	1.418	7.239
120.0	1.985	1.060×10^{-3}	0.892	503.5	2529	503.7	2706	1.528	7.130
130.0	2.701	1.069×10^{-3}	0.669	546.0	2540	546.3	2720	1.634	7.027
140.0	3.613	1.080×10^{-3}	0.509	588.7	2550	589.1	2734	1.739	6.930
150.0	4.758	1.091×10^{-3}	0.393	631.7	2559	632.2	2746	1.842	6.838
160.0	6.178	1.102×10^{-3}	0.307	674.9	2568	675.5	2758	1.943	6.750
170.0	7.916	1.114×10^{-3}	0.243	718.3	2576	719.2	2769	2.042	6.666
180.0	10.02	1.127×10^{-3}	0.194	762.1	2584	763.2	2778	2.140	6.586
190.0	12.54	1.141×10^{-3}	0.157	806.2	2589	807.6	2786	2.236	6.508
200.0	15.54	1.156×10^{-3}	0.127	850.6	2596	852.4	2793	2.331	6.432
210.0	19.06	1.172×10^{-3}	0.104	895.5	2600	897.8	2798	2.425	6.358
220.0	23.18	0.001190	0.086	940.8	2603	943.6	2802	2.518	6.286
230.0	27.95	0.001209	0.072	986.7	2603	990.1	2804	2.610	6.215
240.0	33.44	0.001229	0.060	1033	2603	1037.3	2804	2.702	6.144
250.0	39.73	0.001251	0.050	1080	2603	1085.3	2802	2.793	6.073
260.0	46.88	0.001275	0.042	1128	2600	1134.4	2797	2.884	6.002
270.0	54.98	0.001302	0.036	1177	2592	1184.5	2790	2.975	5.930
280.0	64.11	0.001332	0.030	1227	2587	1236.0	2780	3.067	5.857
290.0	74.36	0.001365	0.026	1279	2573	1289.0	2766	3.159	5.782
300.0	85.81	0.001403	0.022	1332	2560	1344.0	2749	3.253	5.704
320.0	112.7	0.001499	0.015	1445	2531	1461.5	2700	3.448	5.536
340.0	145.9	0.001638	0.011	1570	2462	1594.1	2622	3.659	5.336
360.0	186.5	0.001893	0.007	1725	2351	1760.5	2481	3.915	5.053
374.14	220.9	0.003155	0.003155	2030	2030	2099.0	2099	4.430	4.430

Saturated Steam Pressure Table

p, bar	T, ^{o}C	v_f, m^3/kg	v_g, m^3/kg	u_f, kJ/kg	u_g, kJ/kg	h_f, kJ/kg	h_g, kJ/kg	s_f, kJ/kg	s_g, kJ/kg
0.02	17.50	1.001×10^{-3}	67.00	73.48	2400	73.48	2534	0.261	8.724
0.04	28.96	1.004×10^{-3}	34.80	121.4	2415	121.4	2554	0.423	8.475
0.06	36.15	1.006×10^{-3}	23.75	151.5	2425	151.5	2567	0.521	8.331
0.08	41.50	1.008×10^{-3}	18.1	173.8	2432	173.8	2577	0.593	8.229
0.1	45.8	1.010×10^{-3}	14.68	191.8	2438	191.8	2585	0.649	8.150
0.2	60.07	1.017×10^{-3}	7.649	251.4	2457	251.4	2610	0.832	7.908
0.3	69.11	1.023×10^{-3}	5.229	289.2	2468	289.2	2625	0.944	7.769
0.4	75.87	1.026×10^{-3}	3.994	317.5	2477	317.6	2637	1.026	7.670
0.5	81.33	1.030×10^{-3}	3.240	340.4	2484	340.5	2646	1.091	7.594
0.6	85.94	1.033×10^{-3}	2.732	359.8	2490	359.9	2653	1.145	7.532
0.8	93.5	1.039×10^{-3}	2.087	391.6	2499	391.7	2666	1.233	7.435
1.0	99.62	1.043×10^{-3}	1.694	417.3	2506	417.4	2675	1.303	7.359
1.5	111.4	1.053×10^{-3}	1.159	466.9	2520	467.1	2694	1.434	7.223
2.0	120.2	1.061×10^{-3}	0.886	504.5	2530	504.7	2707	1.530	7.127
3.0	133.6	1.073×10^{-3}	0.606	561.1	2544	561.5	2725	1.672	6.992
4.0	143.6	1.084×10^{-3}	0.463	604.3	2554	604.8	2739	1.777	6.896
5.0	151.9	1.093×10^{-3}	0.375	639.7	2561	640.2	2749	1.861	6.821
6.0	158.9	1.101×10^{-3}	0.316	669.9	2567	670.6	2757	1.931	6.760
7.0	165.0	1.108×10^{-3}	0.273	696.4	2573	697.2	2764	1.992	6.708
8.0	170.4	1.115×10^{-3}	0.240	720.2	2577	721.1	2769	2.046	6.663
10.0	179.9	1.127×10^{-3}	0.194	761.7	2584	762.8	2778	2.139	6.586
20.0	212.4	1.177×10^{-3}	0.100	906.4	2600	908.8	2800	2.447	6.341
30.0	233.9	1.217×10^{-3}	0.067	1005	2604	1008	2804	2.646	6.187
40.0	250.4	1.252×10^{-3}	0.050	1082	2602	1087	2801	2.796	6.070
50.0	264.0	1.286×10^{-3}	0.039	1148	2597	1154	2794	2.920	5.973
60.0	275.6	1.319×10^{-3}	0.032	1205	2590	1213	2784	3.027	5.889
70.0	285.9	1.352×10^{-3}	0.027	1258	2580	1267	2772	3.121	5.813
80.0	295.1	1.384×10^{-3}	0.024	1306	2570	1317	2758	3.207	5.743
90.0	303.4	1.418×10^{-3}	0.021	1350	2558	1363	2742	3.286	5.677
100.0	311.1	1.453×10^{-3}	0.018	1393	2545	1408	2725	3.360	5.614
120.0	324.8	0.001527	0.014	1473	2513	1491	2685	3.496	5.492
140.0	336.8	0.001611	0.012	1549	2477	1571	2638	3.623	5.372
160.0	347.4	0.001711	0.009	1623	2432	1650	2581	3.746	5.246
180.0	357.0	0.001839	0.008	1699	2375	1732	2510	3.871	5.105
200.0	365.8	0.002036	0.006	1786	2295	1826	2411	4.013	4.931
220.9	374.1	0.003155	0.003155	2030	2030	2099	2099	4.430	4.430

Superheated Steam Tables

Pressure p = 0.1 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
50.0	14.87	2444	2593	8.175
100.0	17.20	2516	2688	8.448
150.0	19.51	2588	2783	8.688
200.0	21.83	2661	2880	8.904
250.0	24.14	2736	2977	9.100
300.0	26.45	2812	3077	9.281
400.0	31.06	2969	3280	9.608
500.0	35.68	3132	3489	9.898
600.0	40.30	3303	3705	10.16
700.0	44.91	3480	3929	10.40
800.0	54.14	3664	4159	10.63

Pressure p = 0.5 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
100.0	3.420	2512	2683	7.695
150.0	3.889	2586	2780	7.940
200.0	4.356	2660	2878	8.158
250.0	4.820	2735	2976	8.356
300.0	5.284	2811	3076	8.537
400.0	6.209	2969	3279	8.864
500.0	7.134	3132	3489	9.155
600.0	8.057	3302	3705	9.418
700.0	8.981	3479	3929	9.660
800.0	9.904	3664	4159	9.885
900.0	10.83	3855	4396	10.10

Pressure p = 1.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
100.0	1.696	2507	2676	7.361
150.0	1.936	2583	2776	7.613
200.0	2.172	2658	2875	7.834
250.0	2.406	2733	2974	8.033
300.0	2.639	2810	3074	8.216
400.0	3.103	2968	3278	8.544
500.0	3.565	3132	3488	8.834
600.0	4.028	3302	3704	9.098
700.0	4.490	3479	3928	9.340
800.0	4.952	3664	4159	9.565
900.0	5.414	3855	4396	9.777

Pressure p = 2.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
150.0	0.960	2577	2769	7.280
200.0	1.080	2654	2871	7.507
250.0	1.199	2731	2971	7.709
300.0	1.316	2809	3072	7.893
400.0	1.549	2967	3277	8.222
500.0	1.781	3131	3487	8.513
600.0	2.013	3301	3704	8.777
700.0	2.244	3479	3928	9.019
800.0	2.475	3663	4158	9.245
900.0	2.705	3855	4396	9.457
1000	2.937	4053	4640	9.656

Pressure p = 3.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
150.0	0.634	2571	2761	7.078
200.0	0.716	2651	2866	7.312
250.0	0.796	2729	2968	7.517
300.0	0.875	2807	3069	7.702
400.0	1.032	2966	3275	8.033
500.0	1.187	3130	3486	8.325
600.0	1.341	3301	3703	8.589
700.0	1.496	3478	3927	8.832
800.0	1.650	3663	4158	9.058
900.0	1.804	3854	4395	9.269
1000	1.958	4052	4640	9.469

Pressure p = 4.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
150.0	0.471	2565	2753	6.930
200.0	0.534	2647	2861	7.171
250.0	0.595	2726	2964	7.379
300.0	0.655	2805	3067	7.566
400.0	0.773	2964	3273	7.899
500.0	0.889	3129	3485	8.191
600.0	1.006	3300	3702	8.456
700.0	1.122	3478	3927	8.699
800.0	1.237	3662	4157	8.924
900.0	1.353	3854	4395	9.136
1000	1.469	4052	4639	9.336

Superheated Steam Tables

Pressure p =5.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
200.0	0.425	2643	2855	7.059
250.0	0.474	2724	2961	7.271
300.0	0.523	2803	3064	7.460
400.0	0.617	2963	3272	7.794
500.0	0.711	3128	3484	8.087
600.0	0.804	3300	3702	8.352
700.0	0.897	3478	3926	8.595
800.0	0.990	3662	4157	8.821
900.0	1.082	3854	4395	9.033
1000	1.175	4052	4639	9.233
1100	1.267	4256	4890	9.422

Pressure p =6.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
200.0	0.352	2639	2850	6.967
250.0	0.394	2721	2957	7.182
300.0	0.434	2801	3062	7.372
400.0	0.514	2962	3270	7.708
500.0	0.592	3128	3483	8.002
600.0	0.670	3299	3701	8.267
700.0	0.747	3477	3925	8.511
800.0	0.825	3662	4157	8.737
900.0	0.902	3853	4394	8.949
1000	0.979	4052	4639	9.149
1100	1.056	4256	4890	9.338

Pressure p = 8.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
200.0	0.261	2631	2839	6.816
250.0	0.293	2716	2950	7.038
300.0	0.324	2797	3057	7.233
400.0	0.384	2960	3267	7.572
500.0	0.443	3126	3481	7.867
600.0	0.502	3298	3699	8.133
700.0	0.560	3476	3924	8.377
800.0	0.618	3661	4156	8.603
900.0	0.676	3853	4394	8.815
1000	0.734	4051	4638	9.015
1100	0.792	4256	4889	9.205

Pressure p =10 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
200.0	0.206	2622	2828	6.694
250.0	0.233	2710	2943	6.925
300.0	0.258	2793	3051	7.123
400.0	0.307	2957	3264	7.465
500.0	0.354	3124	3479	7.762
600.0	0.401	3297	3698	8.029
700.0	0.448	3475	3923	8.273
800.0	0.494	3660	4155	8.500
900.0	0.541	3852	4393	8.712
1000	0.587	4051	4638	8.912
1100	0.634	4255	4889	9.102

Pressure p =12 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
200.0	0.169	2613	2816	6.590
250.0	0.192	2704	2935	6.829
300.0	0.214	2789	3046	7.032
400.0	0.255	2955	3261	7.377
500.0	0.295	3123	3476	7.676
600.0	0.334	3296	3696	7.944
700.0	0.373	3474	3922	8.188
800.0	0.412	3660	4154	8.415
900.0	0.451	3852	4392	8.627
1000	0.489	4050	4637	8.827
1100	0.528	4255	4888	9.017

Pressure p = 16 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
250.0	0.142	2692	2919	6.673
300.0	0.159	2781	3035	6.884
400.0	0.190	2950	3254	7.237
500.0	0.220	3120	3472	7.539
600.0	0.250	3293	3693	7.808
700.0	0.279	3473	3920	8.054
800.0	0.309	3658	4152	8.281
900.0	0.338	3851	4391	8.494
1000	0.367	4049	4636	8.694
1100	0.396	4254	4887	8.884
1200	0.425	4464	5144	9.064

Superheated Steam Tables

Pressure p = 20 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
250.0	0.111	2680	2903	6.545
300.0	0.125	2773	3024	6.766
400.0	0.151	2945	3248	7.127
500.0	0.176	3116	3468	7.432
600.0	0.200	3291	3690	7.702
700.0	0.223	3471	3917	7.949
800.0	0.247	3657	4150	8.177
900.0	0.270	3849	4389	8.390
1000	0.293	4048	4635	8.590
1100	0.317	4253	4886	8.780
1200	0.340	4463	5143	8.961

Pressure p =30 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
250.0	0.071	2644	2856	6.287
300.0	0.081	2750	2994	6.539
400.0	0.099	2933	3231	6.921
500.0	0.116	3108	3457	7.234
600.0	0.132	3285	3682	7.509
700.0	0.148	3467	3912	7.757
800.0	0.164	3654	4146	7.986
900.0	0.180	3847	4386	8.200
1000	0.195	4045	4632	8.401
1100	0.211	4250	4883	8.591
1200	0.227	4461	5141	8.772

Pressure p = 40 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
300.0	0.059	2725	2961	6.362
400.0	0.073	2920	3214	6.769
500.0	0.086	3100	3445	7.090
600.0	0.099	3279	3674	7.369
700.0	0.111	3462	3906	7.620
800.0	0.123	3650	4142	7.850
900.0	0.135	3844	4382	8.065
1000	0.146	4043	4629	8.266
1100	0.158	4248	4881	8.457
1200	0.170	4459	5138	8.638

Pressure p = 60 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
300.0	0.036	2667	2884	6.067
400.0	0.047	2893	3177	6.541
500.0	0.057	3082	3422	6.880
600.0	0.065	3267	3658	7.168
700.0	0.074	3453	3894	7.423
800.0	0.082	3643	4133	7.657
900.0	0.090	3838	4375	7.873
1000	0.097	4038	4623	8.075
1100	0.105	4243	4875	8.266
1200	0.113	4454	5133	8.447

Pressure p =80 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
300.0	0.024	2591	2785	5.791
400.0	0.034	2864	3138	6.363
500.0	0.042	3064	3398	6.724
600.0	0.048	3254	3642	7.021
700.0	0.055	3444	3882	7.281
800.0	0.061	3636	4124	7.517
900.0	0.067	3832	4368	7.735
1000	0.073	4033	4617	7.938
1100	0.079	4239	4870	8.130
1200	0.085	4450	5129	8.311

Pressure p = 100 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg
350.0	0.022	2699	2923	5.944
400.0	0.026	2832	3097	6.212
500.0	0.033	3046	3374	6.597
600.0	0.038	3242	3625	6.903
700.0	0.044	3435	3871	7.169
800.0	0.049	3629	4115	7.408
900.0	0.053	3826	4361	7.627
1000	0.058	4028	4611	7.832
1100	0.063	4234	4865	8.024
1200	0.069	4445	5124	8.206